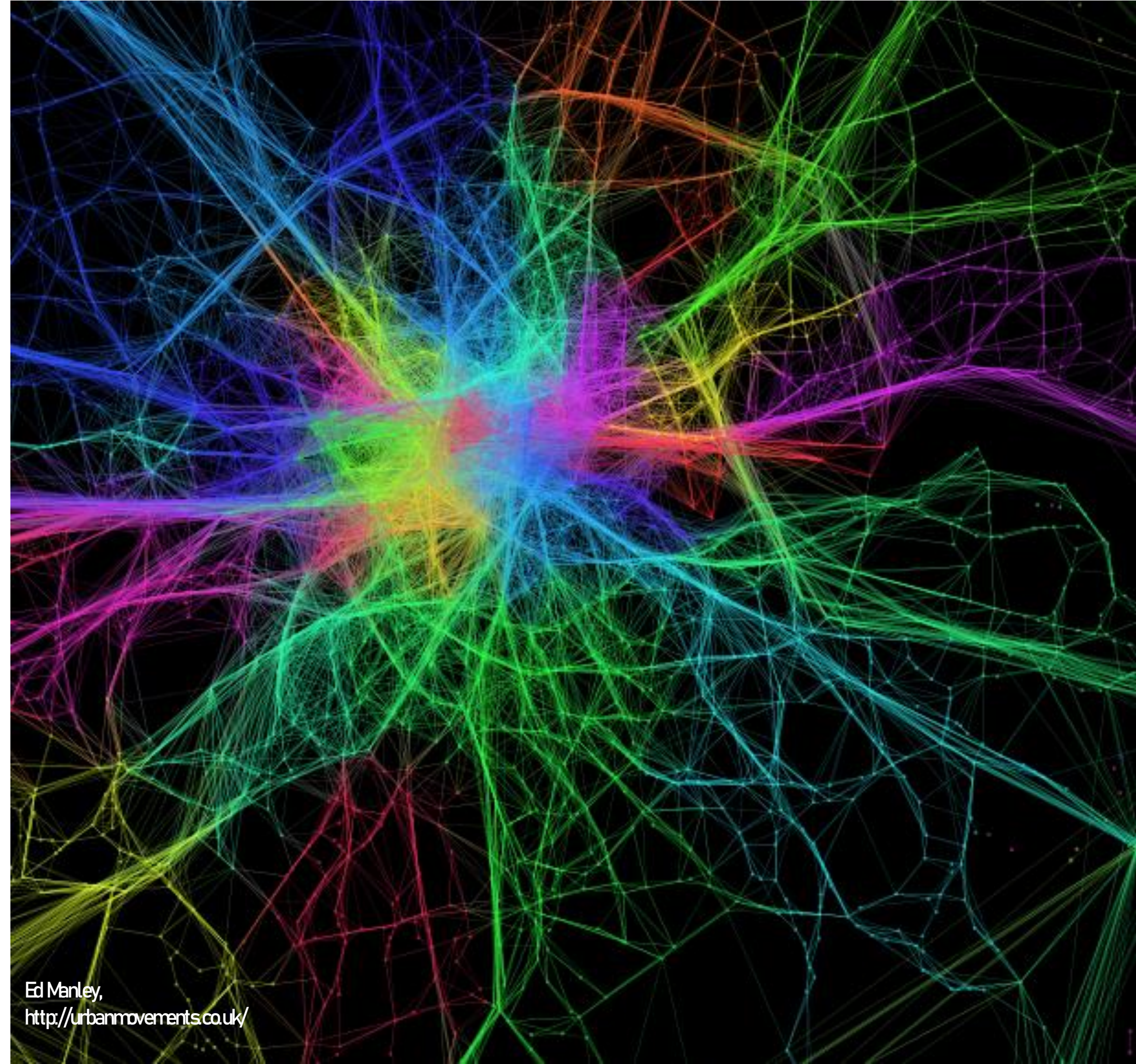


OpenStreetMap and (Spatial) Networks

ENVS456 – week 7
Gabriele Filomena



Ed Manley,
<http://urbanmovements.co.uk/>

Agenda

- **OpenStreetMap:**
 - History and Quality
 - Accessing Data
- **Networks**
 - Graph theory: Basic concepts
 - Graph theory: Centrality measures
 - Street network analysis: representations and applications

OpenStreetMap

History and Quality

Readapted from Michael Szell's slides
see https://github.com/mszell/geospatialdatascience/tree/main/unit08_openstreetmap/materials

OpenStreetMap (OSM) is like Wikipedia for maps

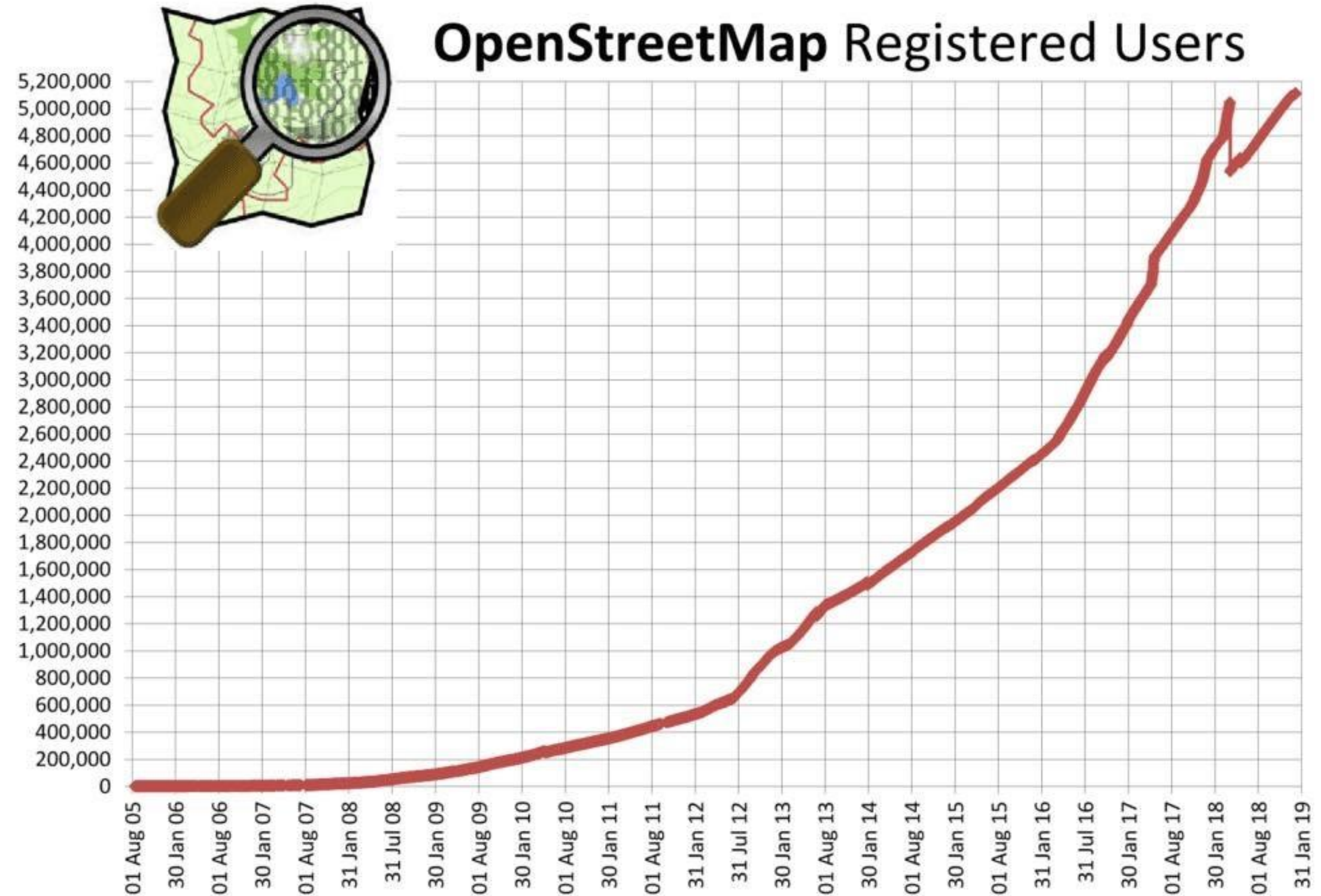
- Established in 2004 by Steve Coast
- Volunteered Geographical Information (VGI)
- The basis for most routing apps and many geo services (Amazon, Mapbox, Tesla, ...)
- Contributors are not just local mappers, but huge organizations



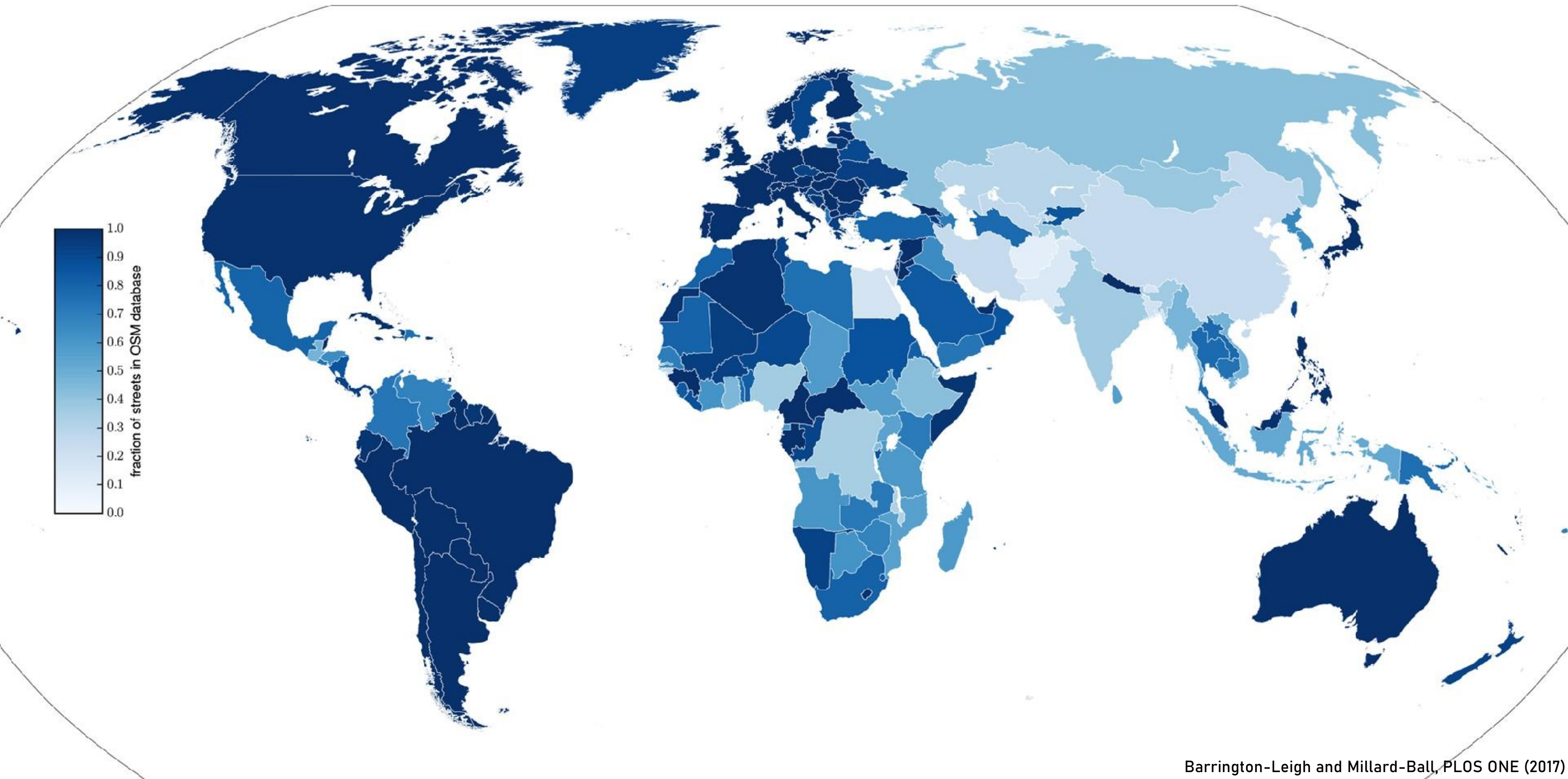
OSM is huge

- 6 Million contributors
- 9 Billion elements

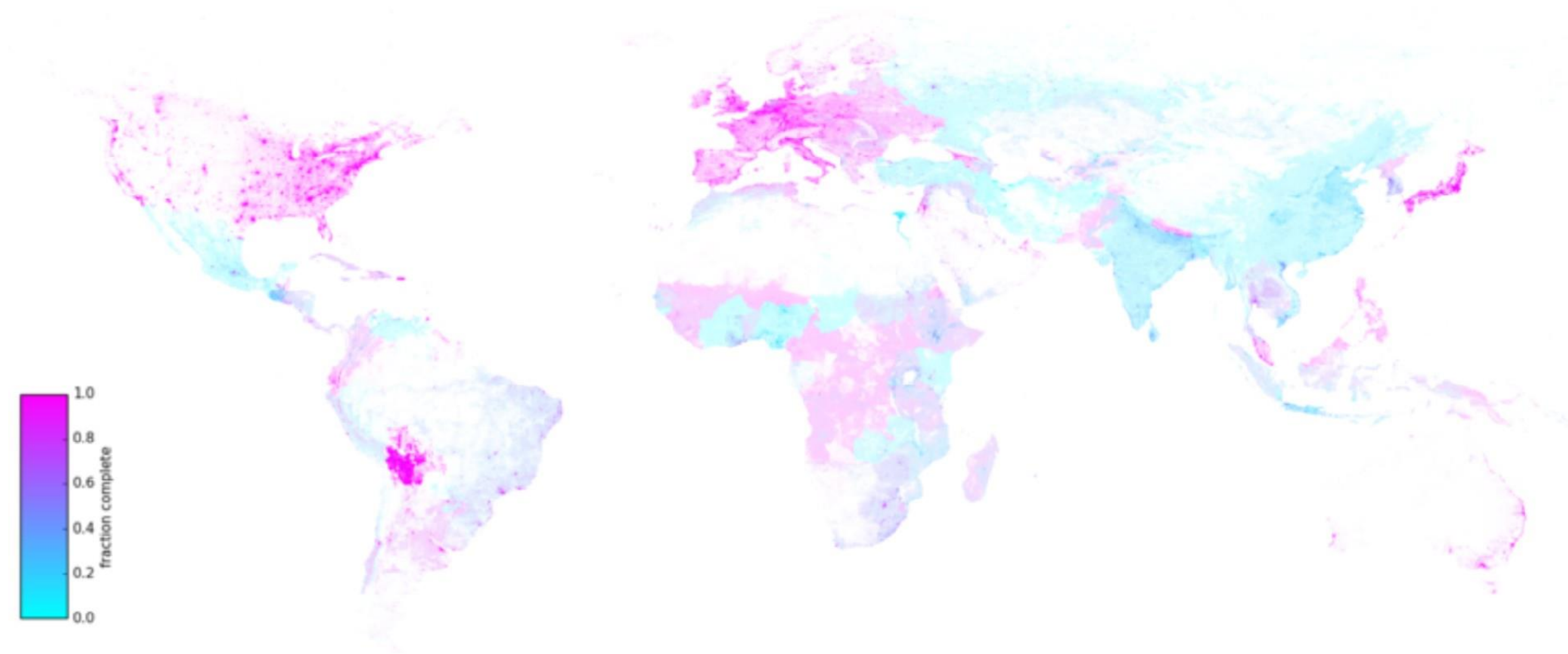
Probably more



OSM is relatively complete

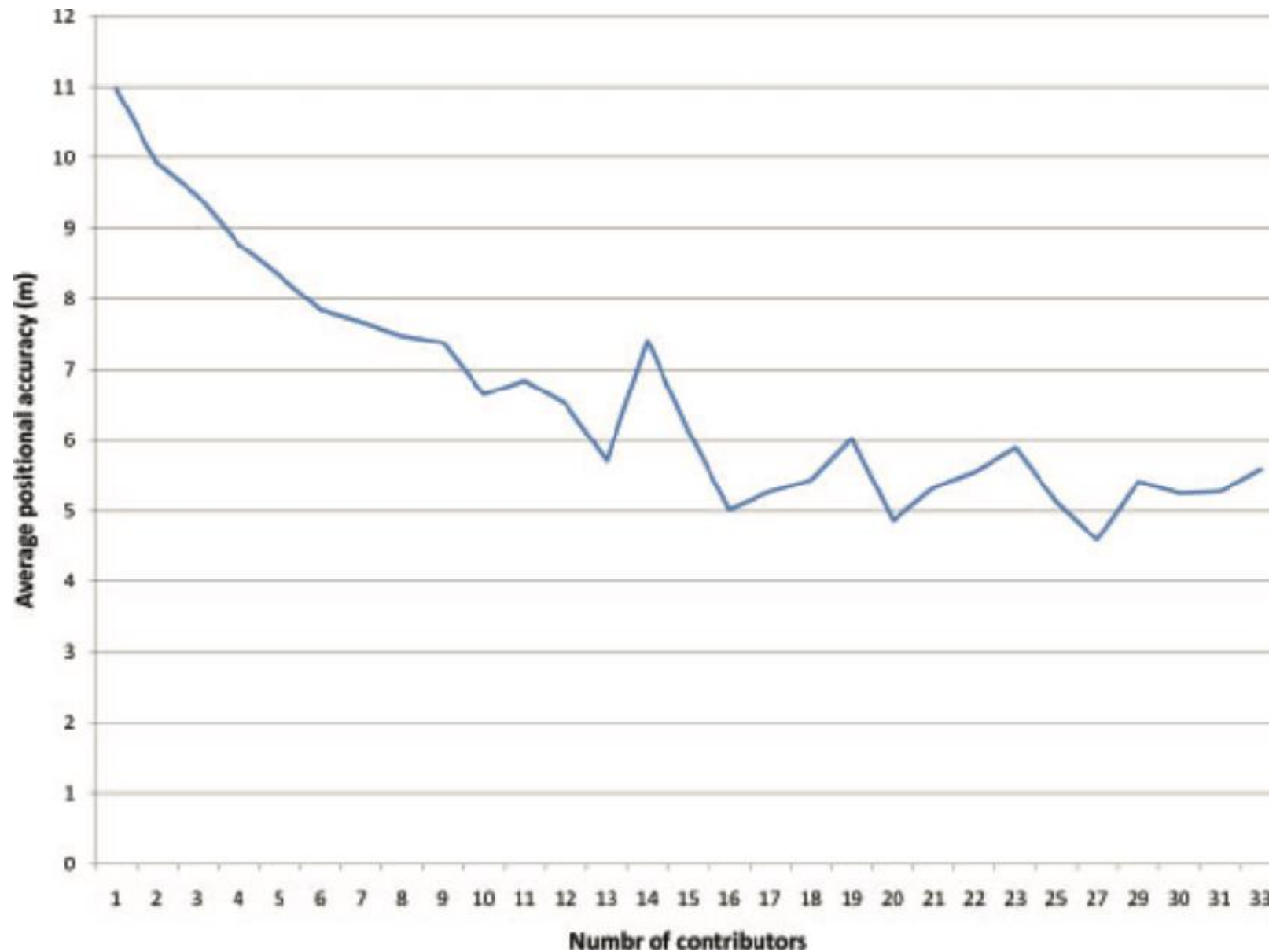


OSM is relatively complete

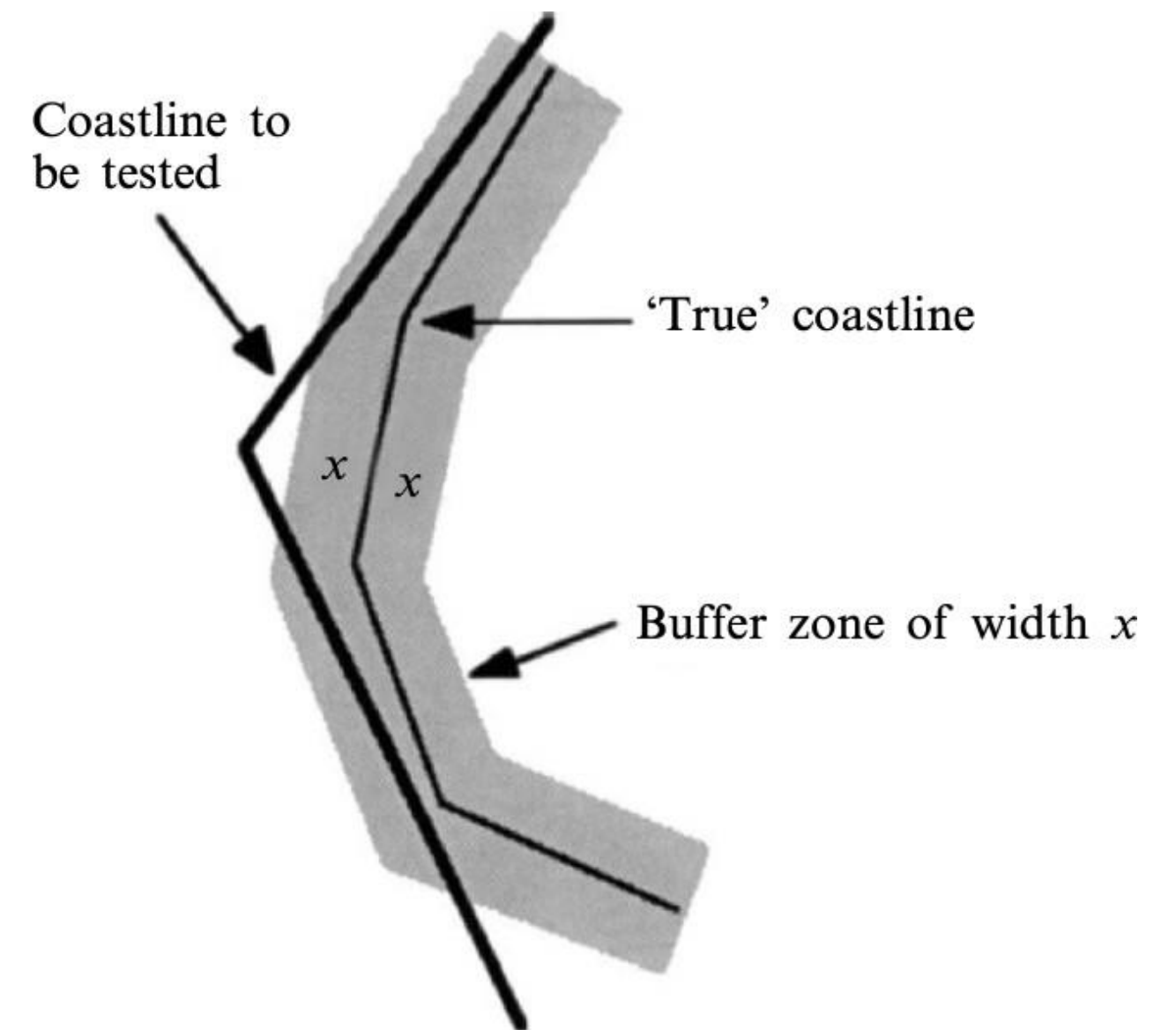


OSM is relatively accurate

The more contributors, the better the accuracy



Buffer-zone method between OSM and ground truth





Why OSM and not Google maps?

OSM

Free, also for commercial applications

Ecosystem of open tools, research

Less polished rendering and UX

Underlies most map-based software

Maps cycle paths, footpaths, etc.

Google maps

Not free, licensing and fees at whim of management

Proprietary

More polished rendering and UX

Integration with Google products

Car-centric, less complete

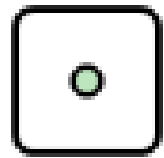
OpenStreetMap

Accessing & Handling OSM Data

Readapted from Michael Szell's slides
see https://github.com/mszell/geospatialdatascience/tree/main/unit08_openstreetmap/materials

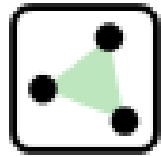
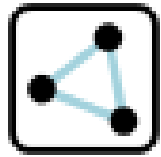
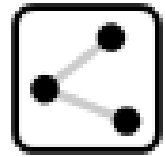
OSM topological data structure

3 elements / primitives:



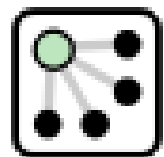
Node

id, lat, lon



Way

ordered list of nodes



Relation

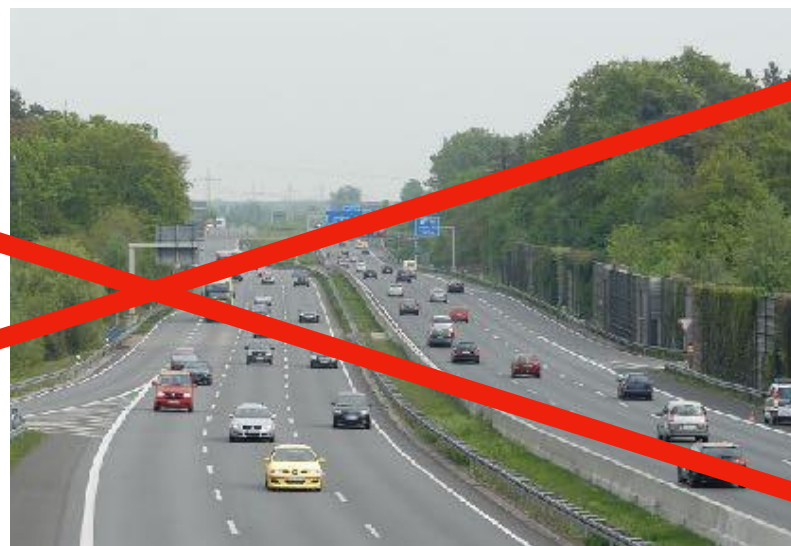
multiple elements



Tag

key=value, *describing an element's feature*

The “highway” tag



How to access OSM Data

- Export from browser.
- Download via Overpass API/turbo .
- Download extracted region from distributor.
- Download via software (like JSOM) -> **AVOID**
- Download via Python (OSMnx) -> **YES**

From browser

 OpenStreetMap

Edit

▼

History

Export

GPS Traces

User Diaries

Copyright

Help

About

 mszell

Search

Where is this?

Go



Export

55.66070

12.58756

12.59531

55.65825

Manually select a different area

Licence

OpenStreetMap data is licensed under the [Open Data Commons Open Database License \(ODbL\)](#).

Export

If the above export fails, please consider using one of the sources listed below:

Overpass API

Download this bounding box from a mirror of the OpenStreetMap database

Planet OSM

Regularly-updated copies of the complete OpenStreetMap database

Geofabrik Downloads

Regularly-updated extracts of continents, countries, and selected cities

Other Sources

Additional sources listed on the OpenStreetMap wiki

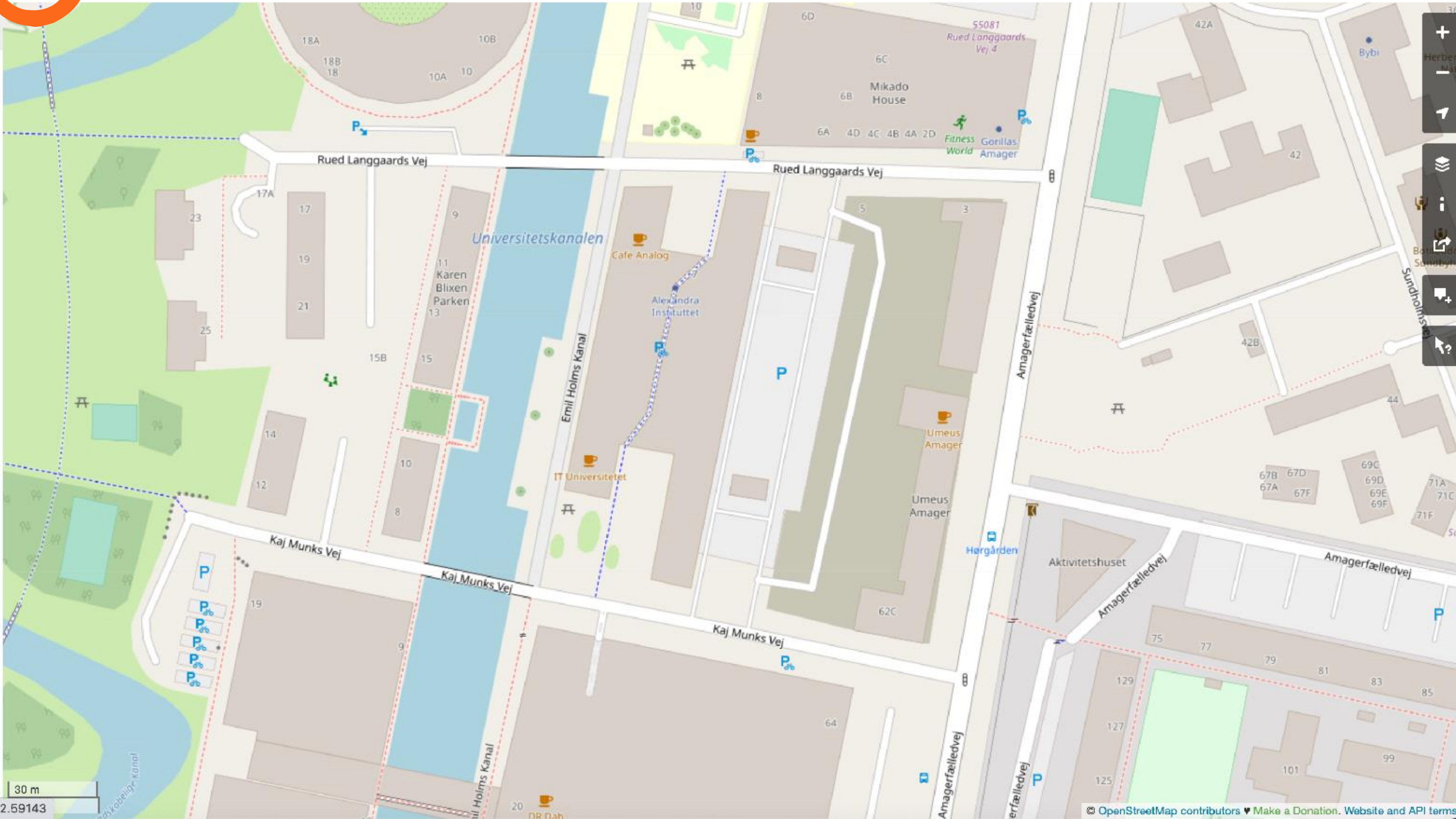
30 m

<https://www.openstreetmap.org/export#map=18/55.65948/12.59143>

© OpenStreetMap contributors

♥ Make a Donation

Website and API terms



From Overpass API

- The Overpass API is a read-only API that serves up parts of the OSM map data.
- It acts as a database over the web: the client sends a query to the API and gets back the data set that corresponds to the query.
- Uses regex, is quite human-unreadable Syntax is limited

See https://wiki.openstreetmap.org/wiki/Overpass_API

```
/*
```

This query looks for nodes, ways and relations with the given key/value combination.

Choose your region and hit the Run button above!

```
*/
```

```
[out:json][timeout:25];
```

```
// gather results
```

```
(
```

```
  // query part for: "amenity=post_box"
```

```
  node[ "amenity"="post_box" ]( {{bbox}} );
```

```
  way[ "amenity"="post_box" ]( {{bbox}} );
```

```
  relation[ "amenity"="post_box" ]( {{bbox}} );
```

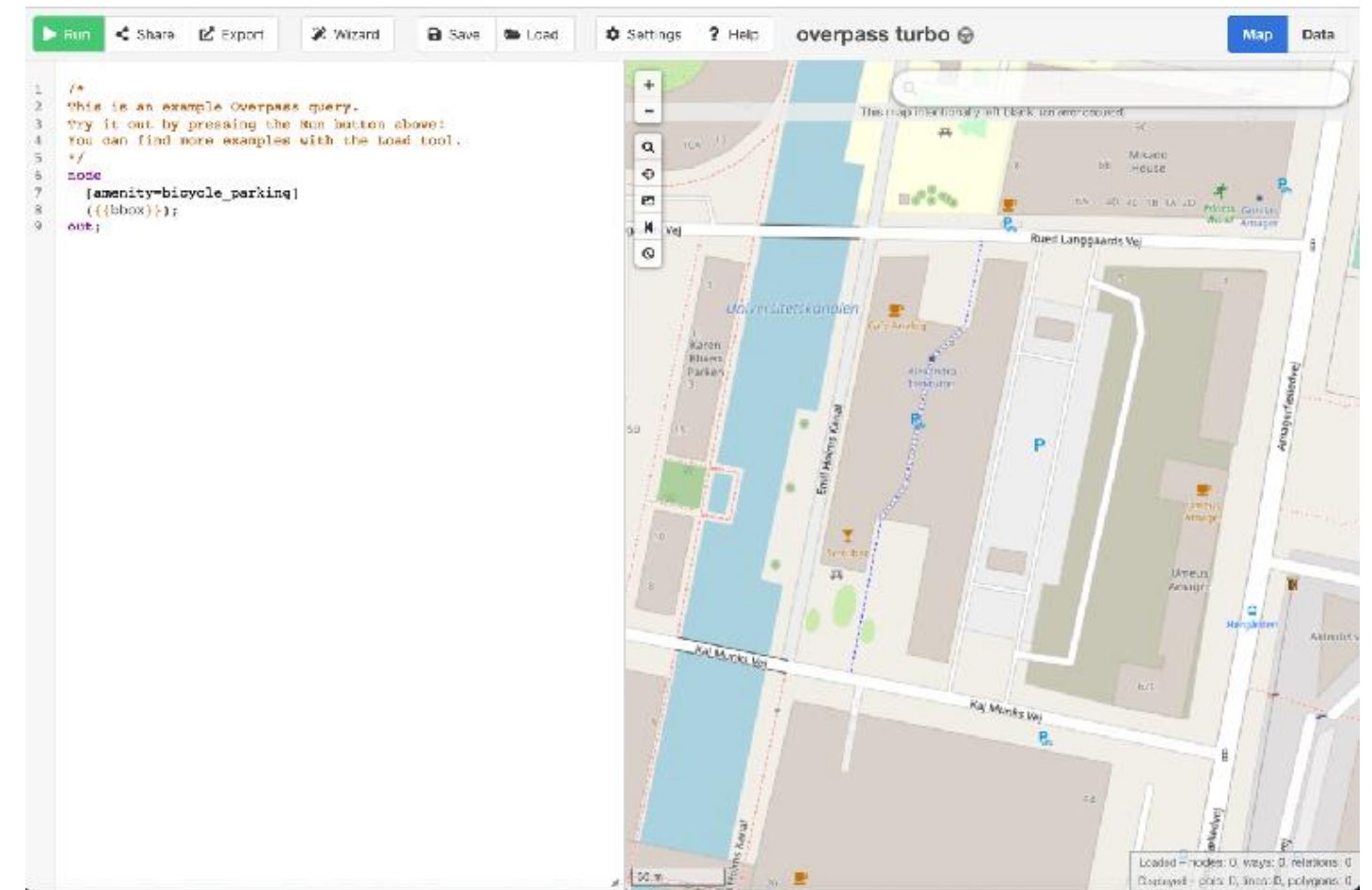
```
);
```

```
// print results
```

```
out body;
```

```
>;
```

```
out skel qt;
```



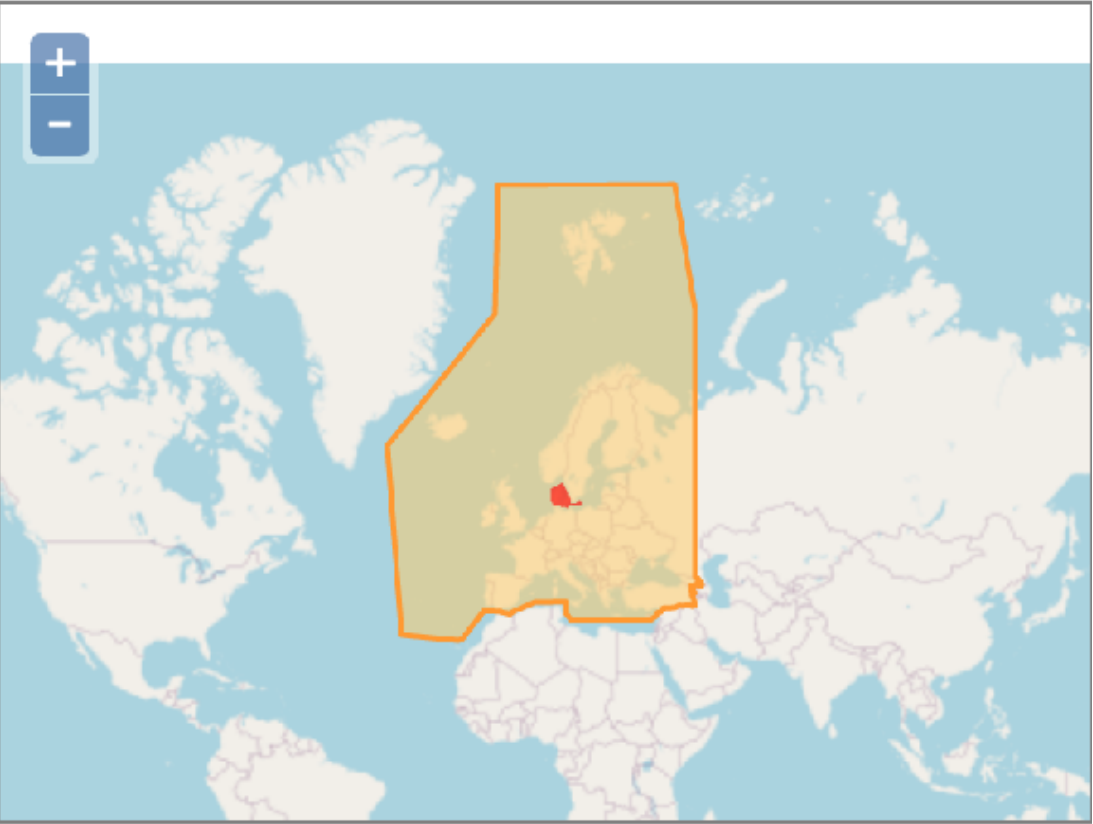
<https://overpass-turbo.eu/>

From Distributors

Sub Regions

Click on the region name to see the overview page for that region, or select one of the file extension links for quick access.

Sub Region	Quick Links		
	.osm.pbf	.shp.zip	.osm.bz2
Albania	[.osm.pbf] (42.7 MB)	[.shp.zip]	[.osm.bz2]
Andorra	[.osm.pbf] (1.8 MB)	[.shp.zip]	[.osm.bz2]
Austria	[.osm.pbf] (644 MB)	[.shp.zip]	[.osm.bz2]
Azores	[.osm.pbf] (12.4 MB)	[.shp.zip]	[.osm.bz2]
Belarus	[.osm.pbf] (244 MB)	[.shp.zip]	[.osm.bz2]
Belgium	[.osm.pbf] (462 MB)	[.shp.zip]	[.osm.bz2]
Bosnia-Herzegovina	[.osm.pbf] (107 MB)	[.shp.zip]	[.osm.bz2]
Bulgaria	[.osm.pbf] (111 MB)	[.shp.zip]	[.osm.bz2]
Croatia	[.osm.pbf] (135 MB)	[.shp.zip]	[.osm.bz2]
Cyprus	[.osm.pbf] (20.0 MB)	[.shp.zip]	[.osm.bz2]
Czech Republic	[.osm.pbf] (758 MB)	[.shp.zip]	[.osm.bz2]
Denmark	[.osm.pbf] (394 MB)	[.shp.zip]	[.osm.bz2]
Estonia	[.osm.pbf] (94 MB)	[.shp.zip]	[.osm.bz2]
Faroe Islands	[.osm.pbf] (4.6 MB)	[.shp.zip]	[.osm.bz2]



<https://download.geofabrik.de/>

Custom extract <https://extract.bbbike.org/>

Via OSMnx (OSM + NetworkX)

 **gboeing / osmnx** Public

 Watch 115  Fork 668  Starred 3.5k

 **Code**  Issues 7  Pull requests 1  Actions  Projects ...

 main Go to file Add file **Code**

 **gboeing** pin black ver... ... ✓ 3 days ago 🕒 2,446

	.github	drop python 3.7 support...	9 days ago
	docs	version bump	4 months ago

About

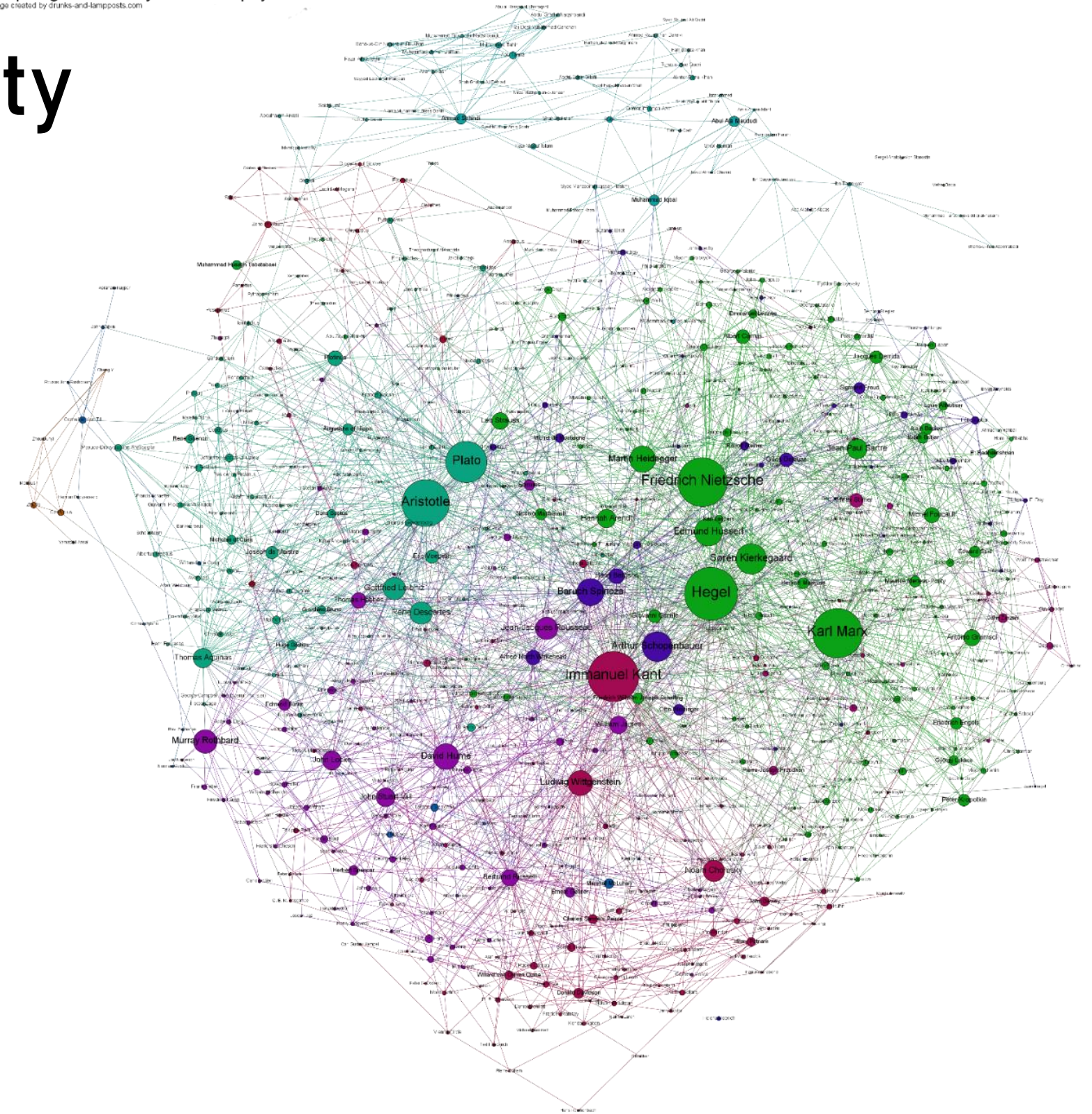
OSMnx: Python for street networks. Retrieve, model, analyze, and visualize street networks and other spatial data from OpenStreetMap.

<https://osmnx.readthedocs.io/en/stable/>

Networks & Graphs

Network science and Complexity

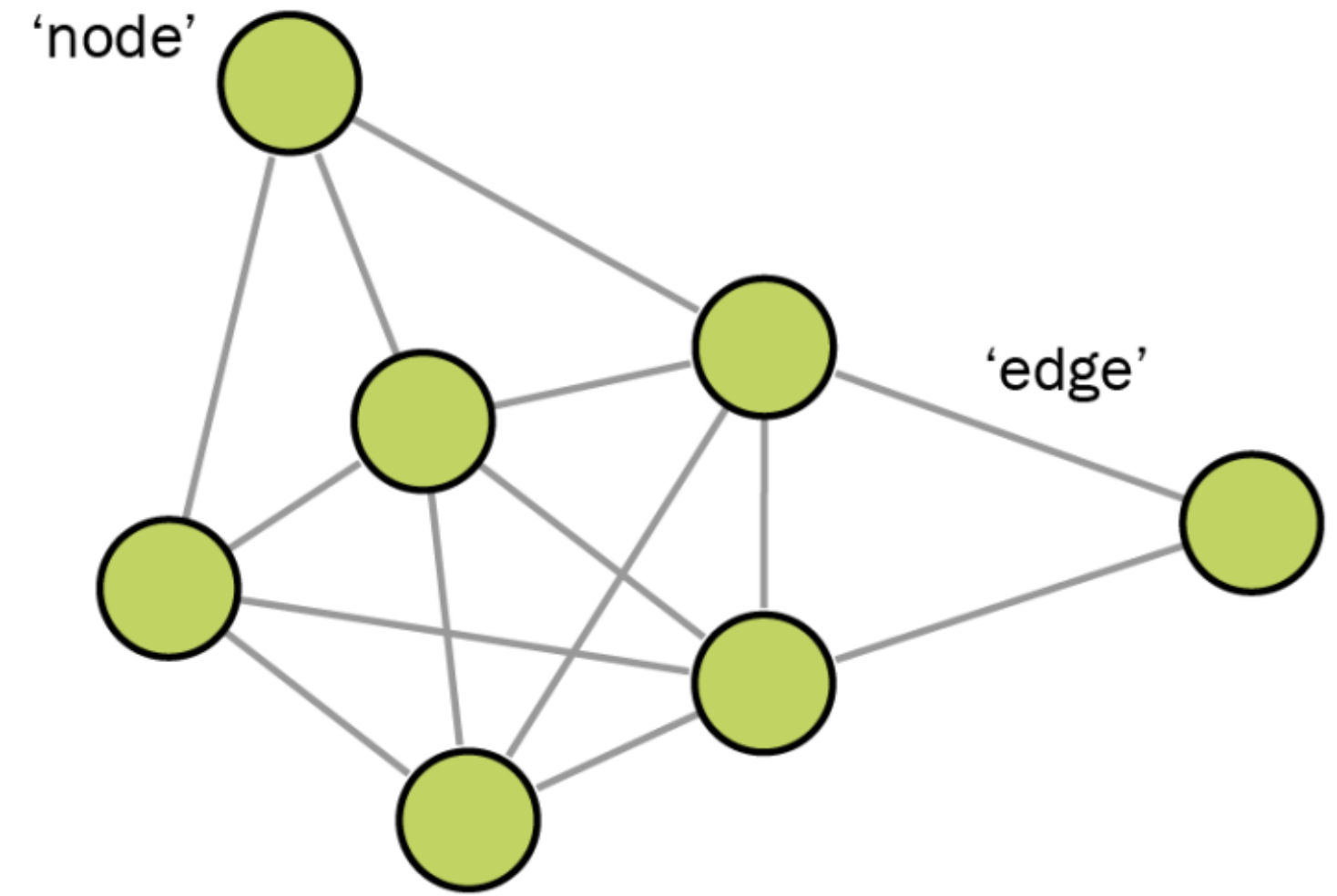
- Complex systems and interaction amongst its parts might be represented in various ways through networks.
- If we understand the dynamics of one system, we can extend the emerging patterns to different systems



Infer generic functioning mechanisms.

Graph Components

- Network science is built upon the foundation of graph theory.
- A *graph* is an abstract mathematical representation of a set of elements and the connections between them.

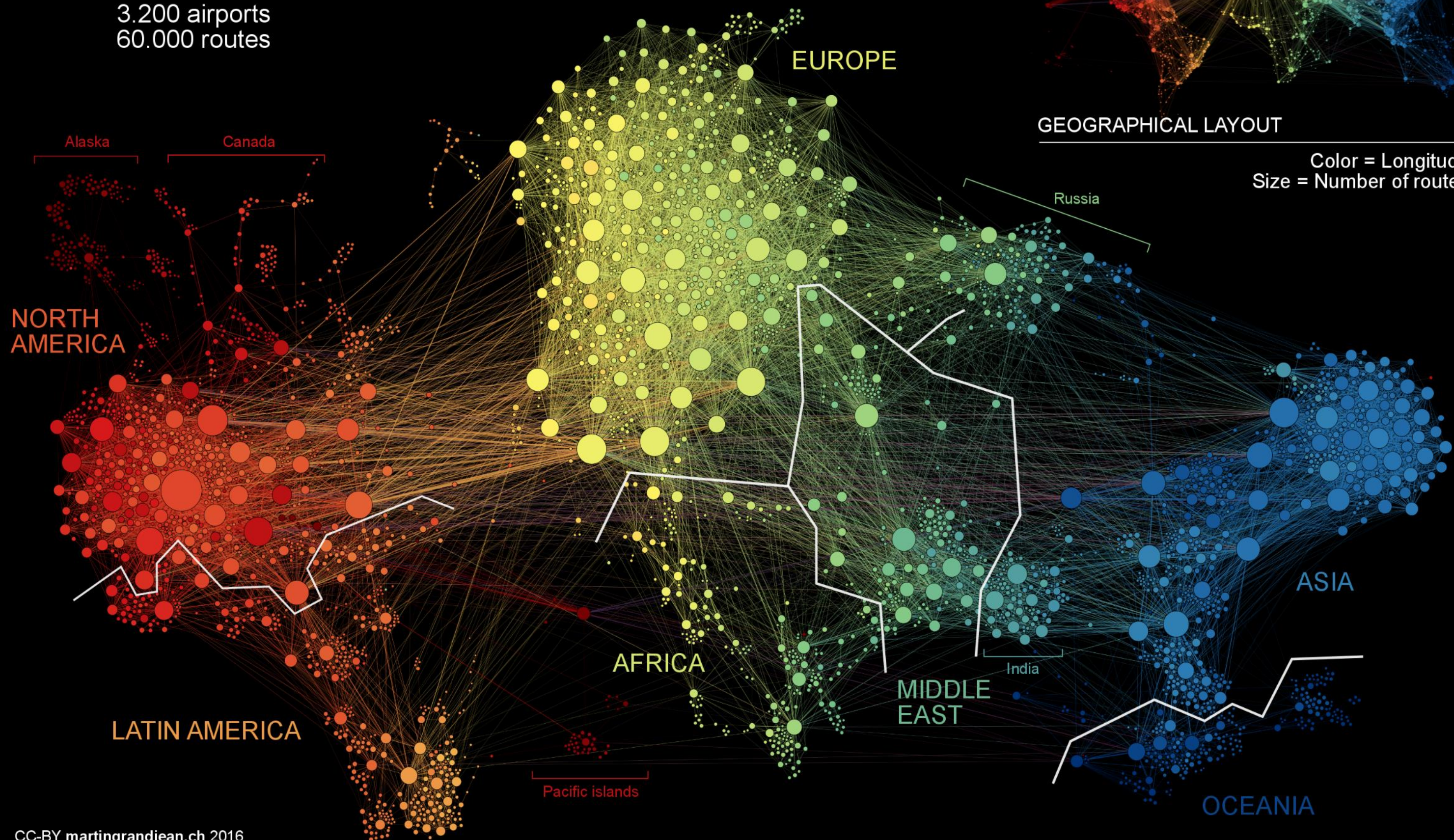


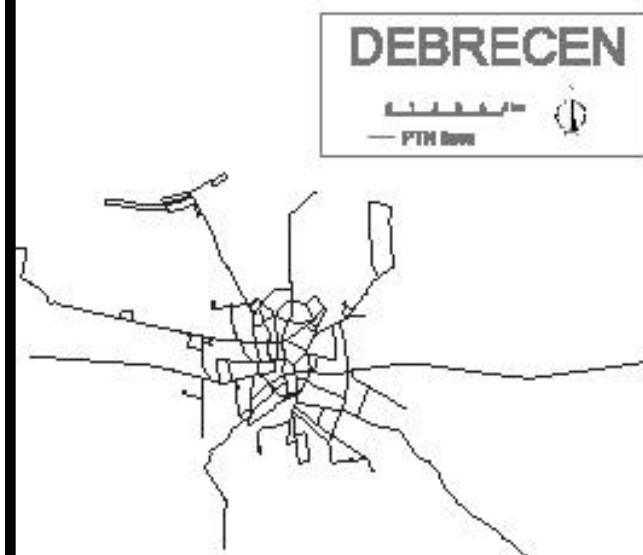
- components: nodes, vertices (n)
- interactions: links, edges (m)
- system: network, graph $G(n, m)$



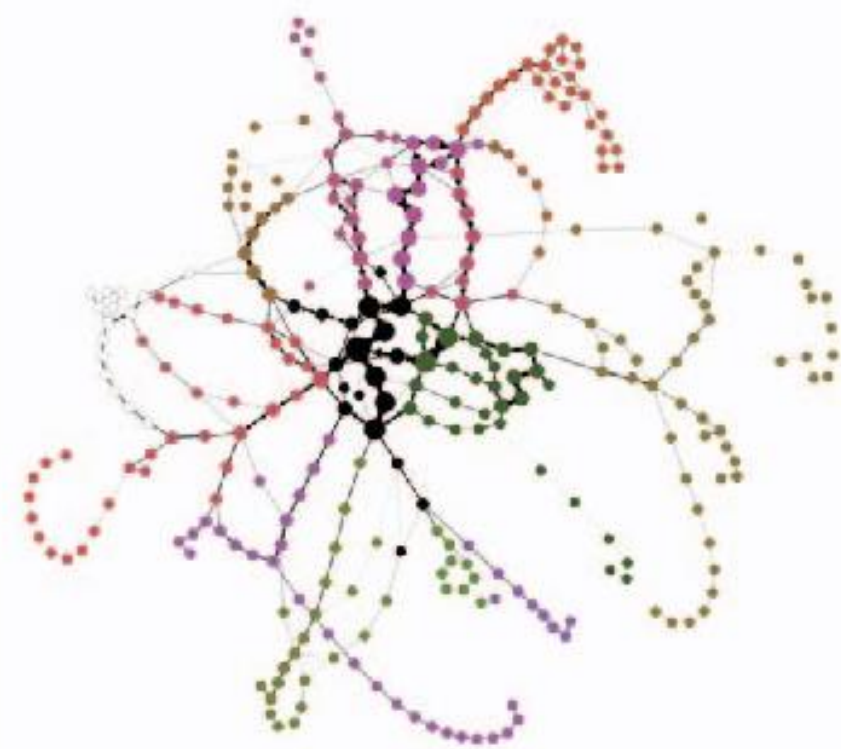
TRANSPORTATION CLUSTERS

3.200 airports
60.000 routes

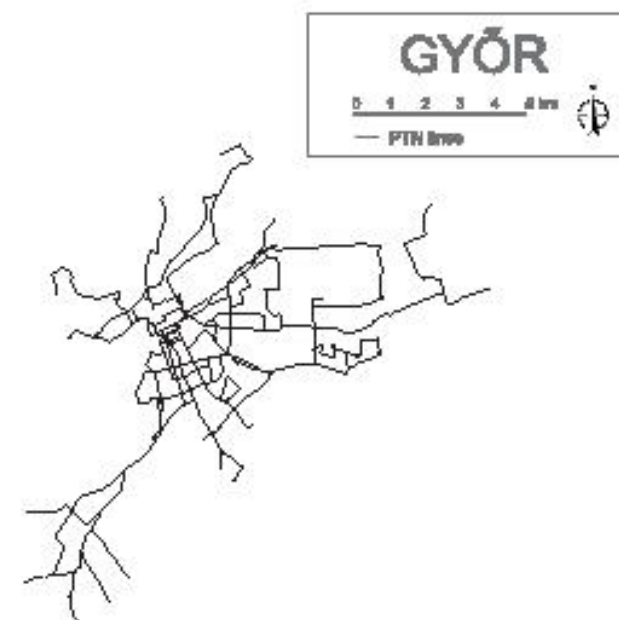




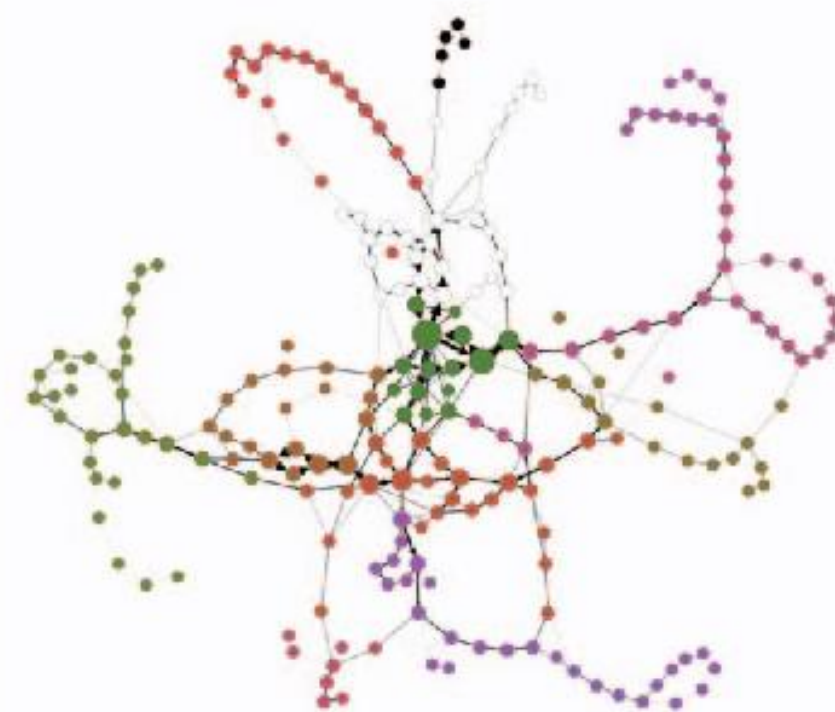
(a)



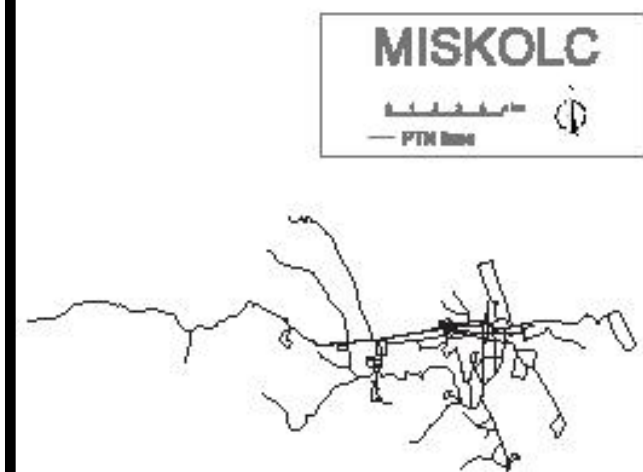
(b)



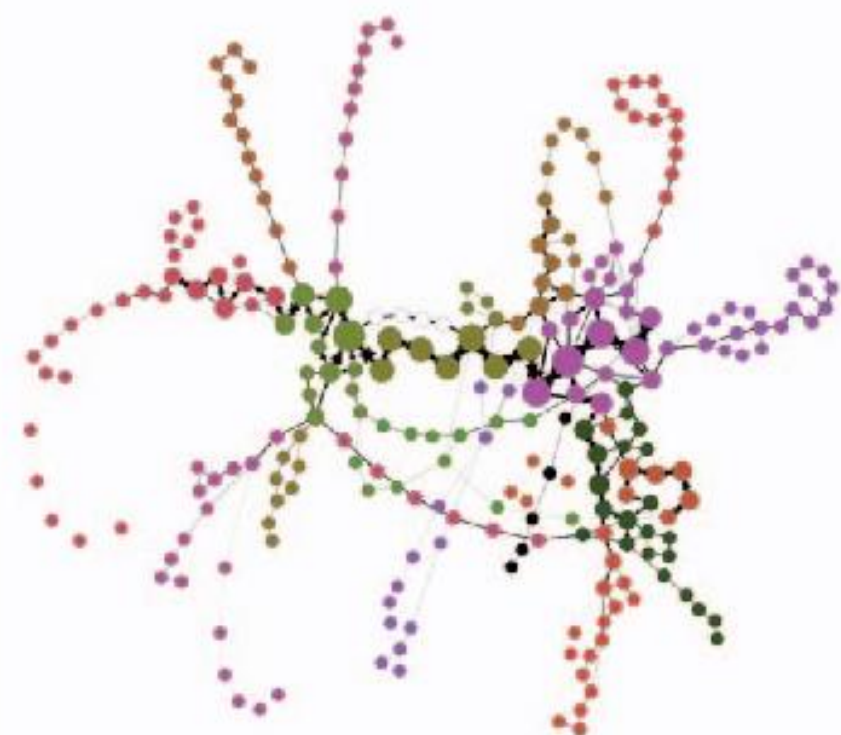
(c)



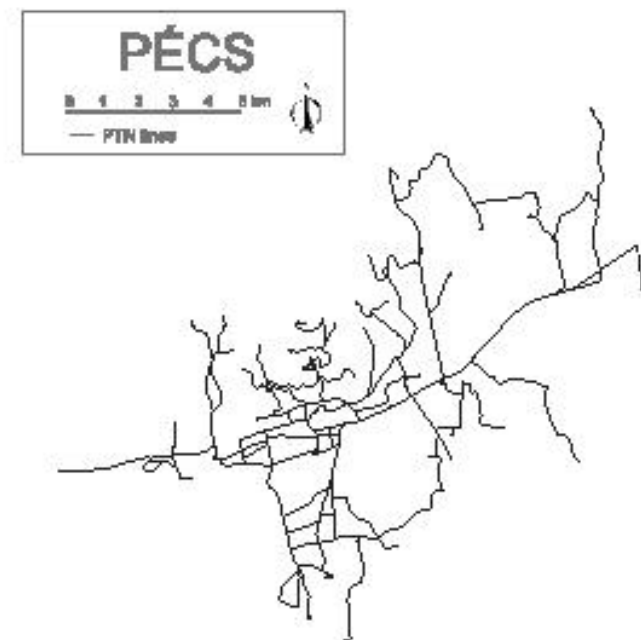
(d)



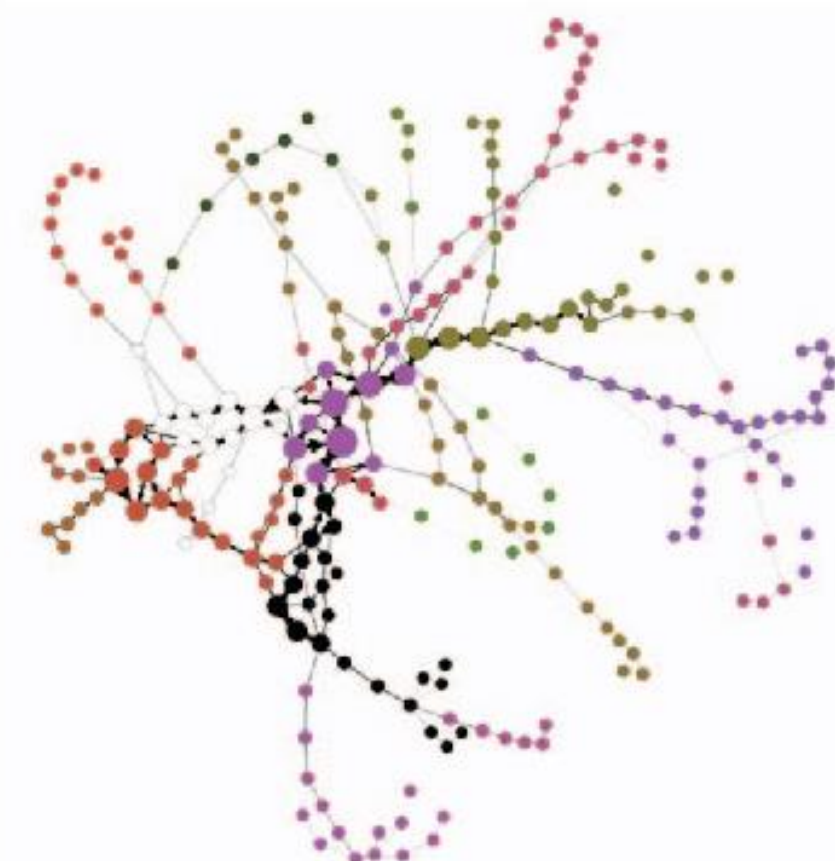
(e)



(f)

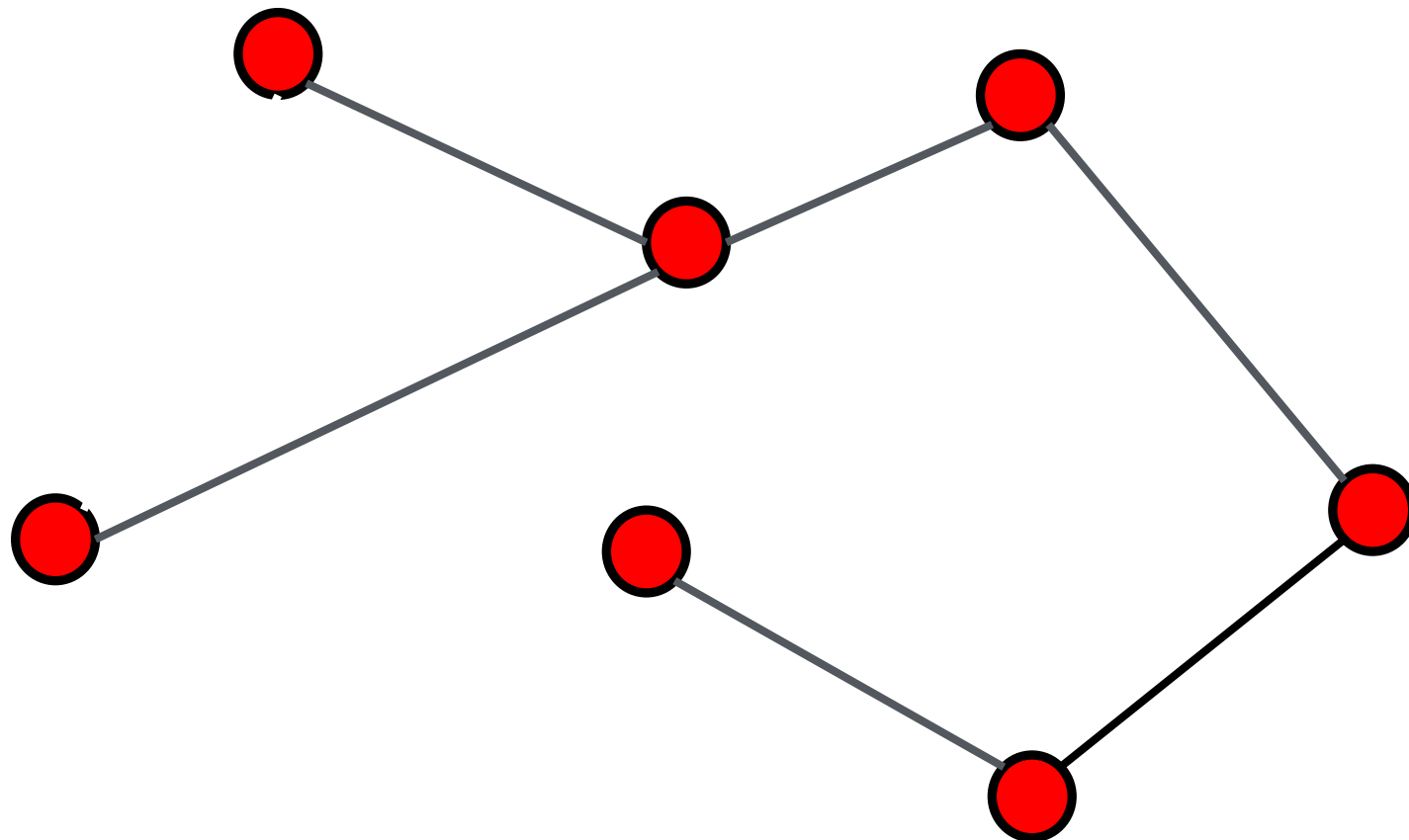


(g)

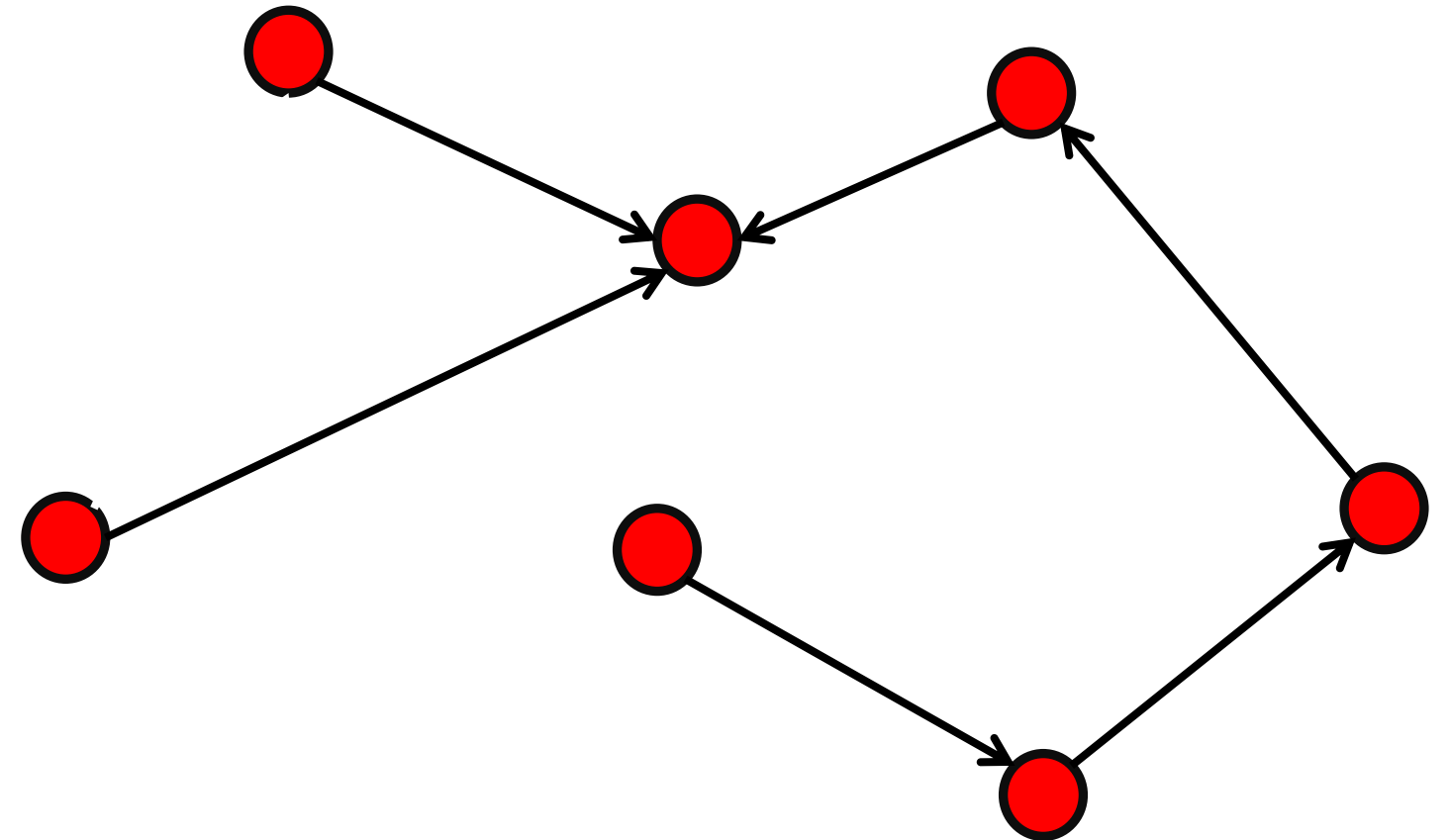


(h)

Undirected graph

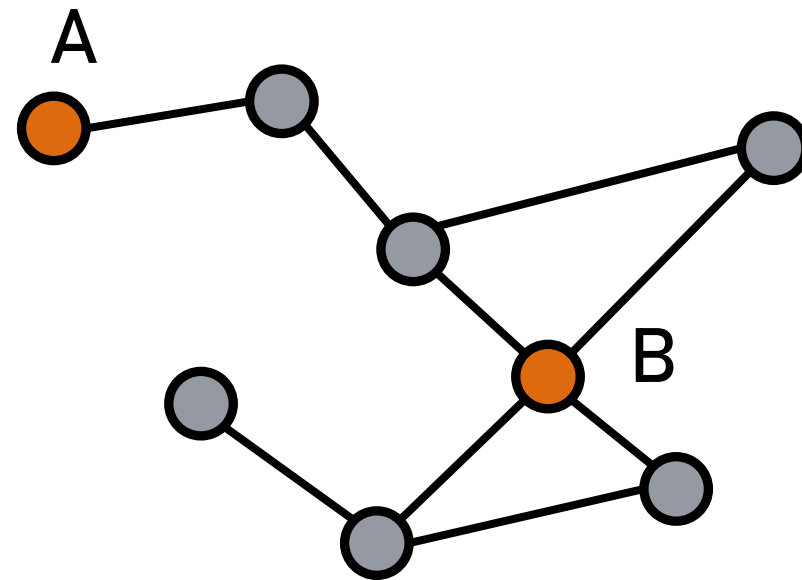


Directed graph (digraph)



Node Degree

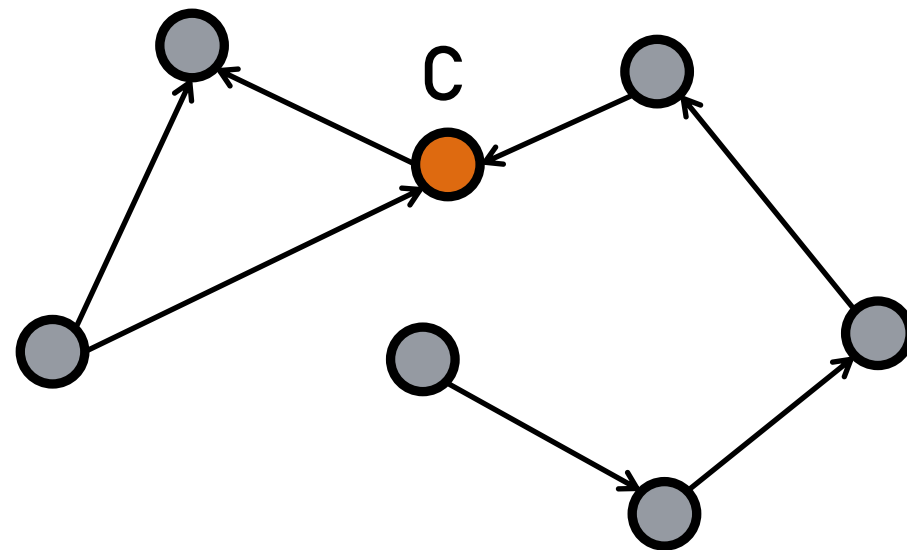
Undirected



The number of links connected to the node

$$k_A = 1 \quad k_B = 4$$

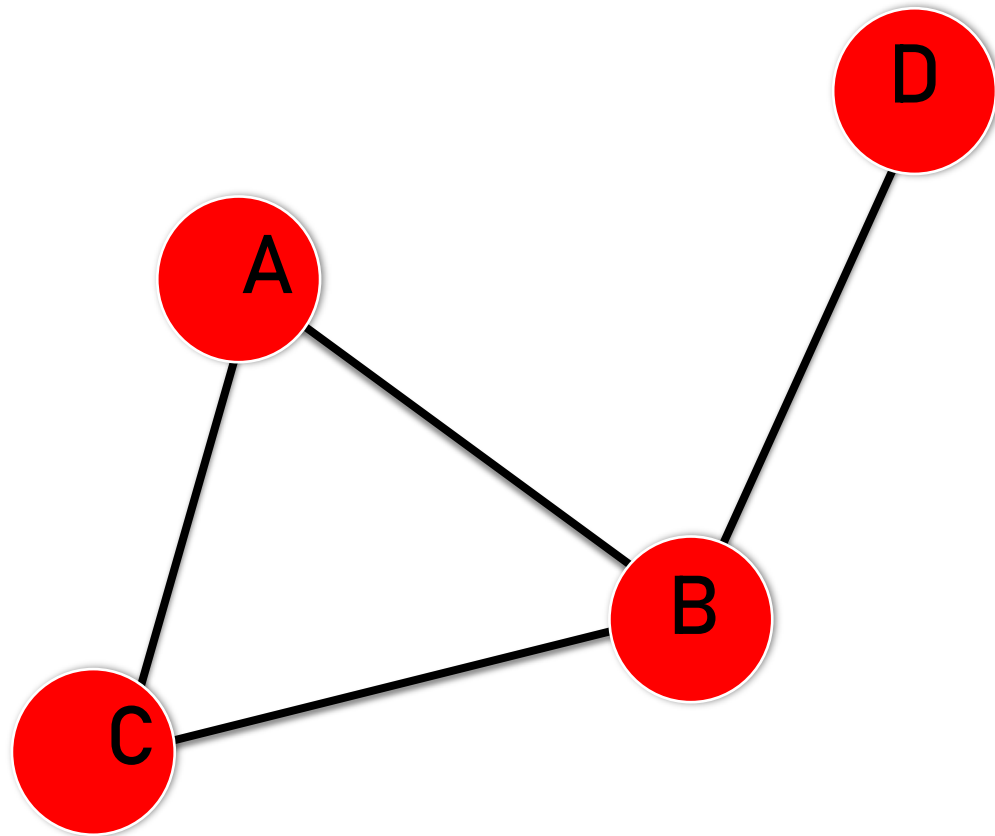
Directed



In directed networks we can define an in-degree and out-degree. The (total) degree is the sum of in- and out-degree.

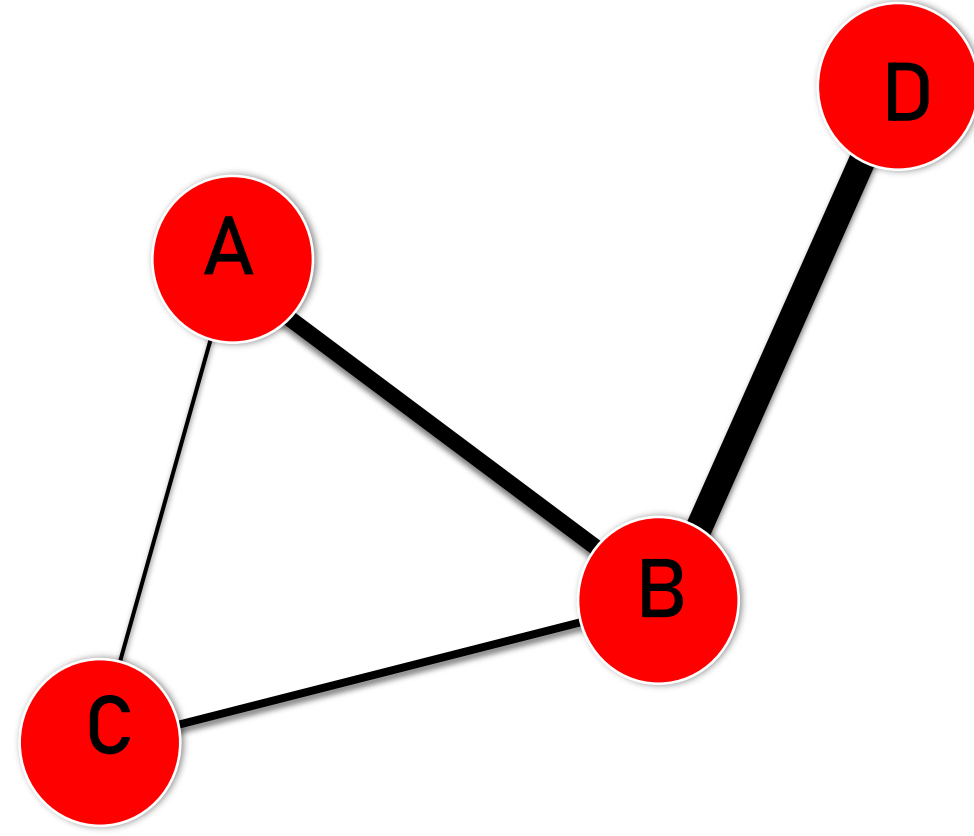
$$k_C^{in} = 2 \quad k_C^{out} = 1 \quad k_C = 3$$

Unweighted Graph



$$A_{ij} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

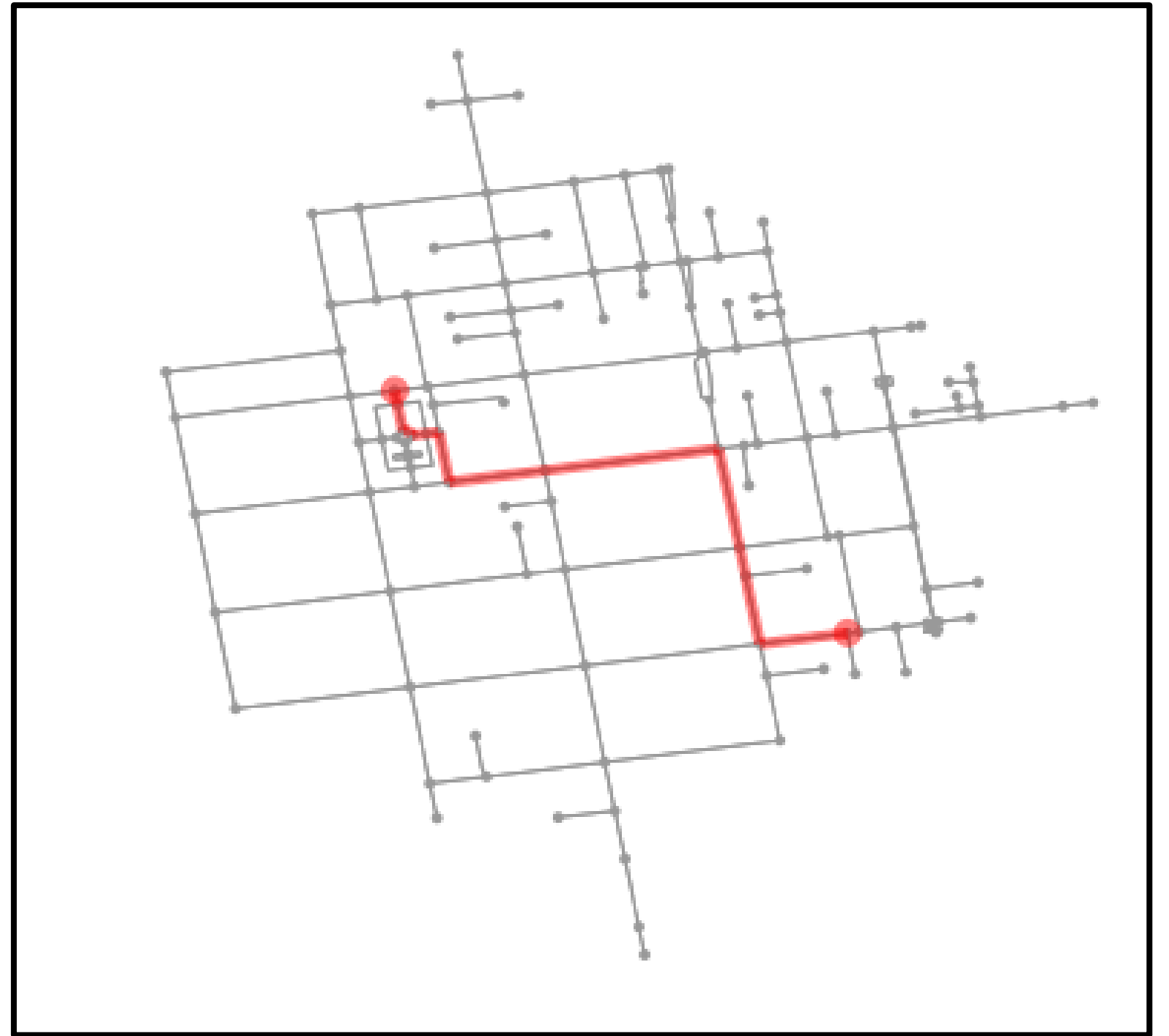
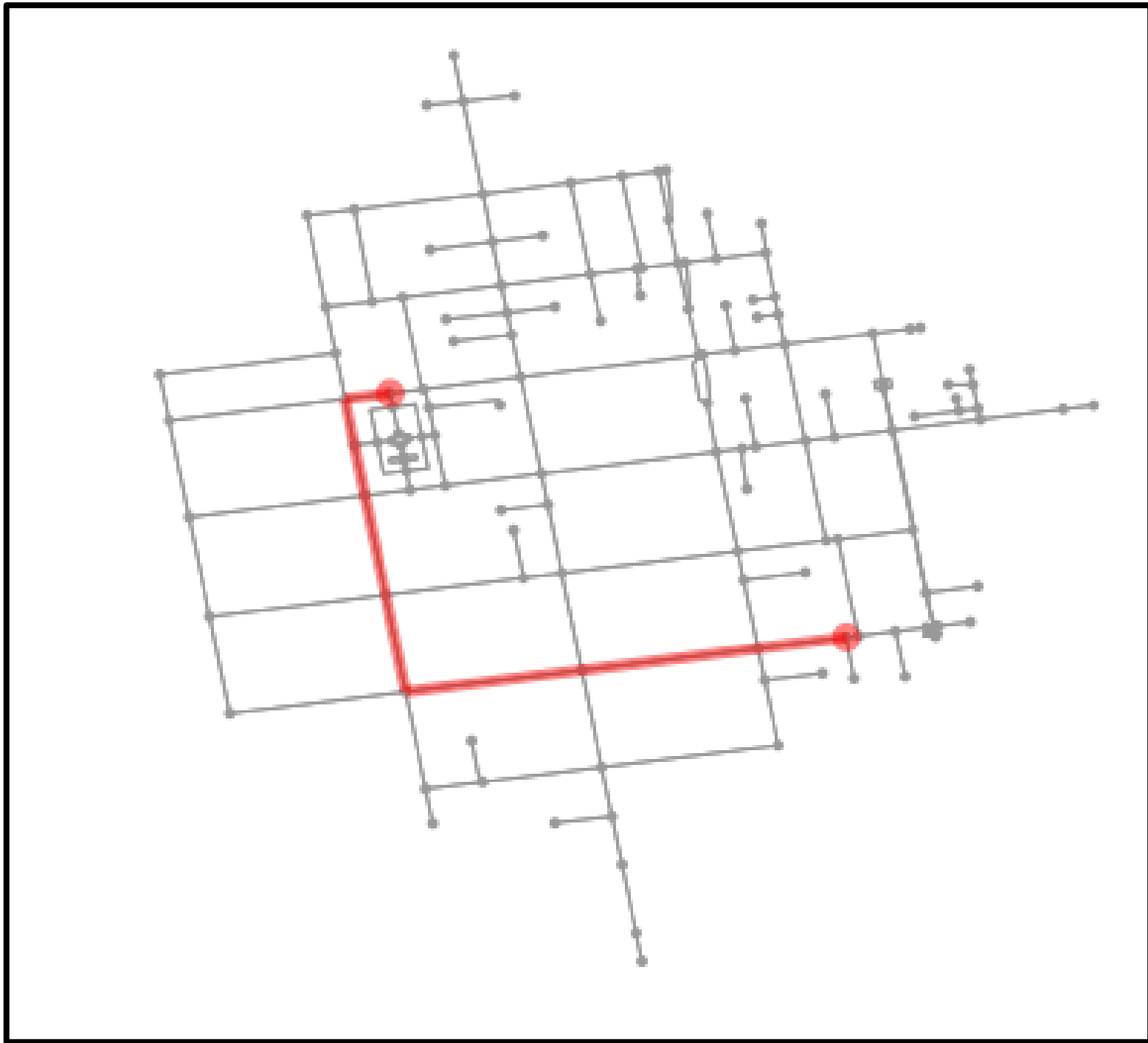
Weighted Graph



$$A_{ij} = \begin{pmatrix} 0 & 2 & 0.5 & 0 \\ 2 & 0 & 1 & 4 \\ 0.5 & 1 & 0 & 0 \\ 0 & 4 & 0 & 0 \end{pmatrix}$$

Distances and shortest path

- The distance between 2 nodes is defined as the number of links along the shortest path connecting them – TOPOLOGICAL DISTANCE
- In *digraphs* each path needs to follow the direction of the arrows. Thus the distance from node A to B may be different from the distance from B to A.
- Graphs representing street networks are usually weighted – ROAD DISTANCE

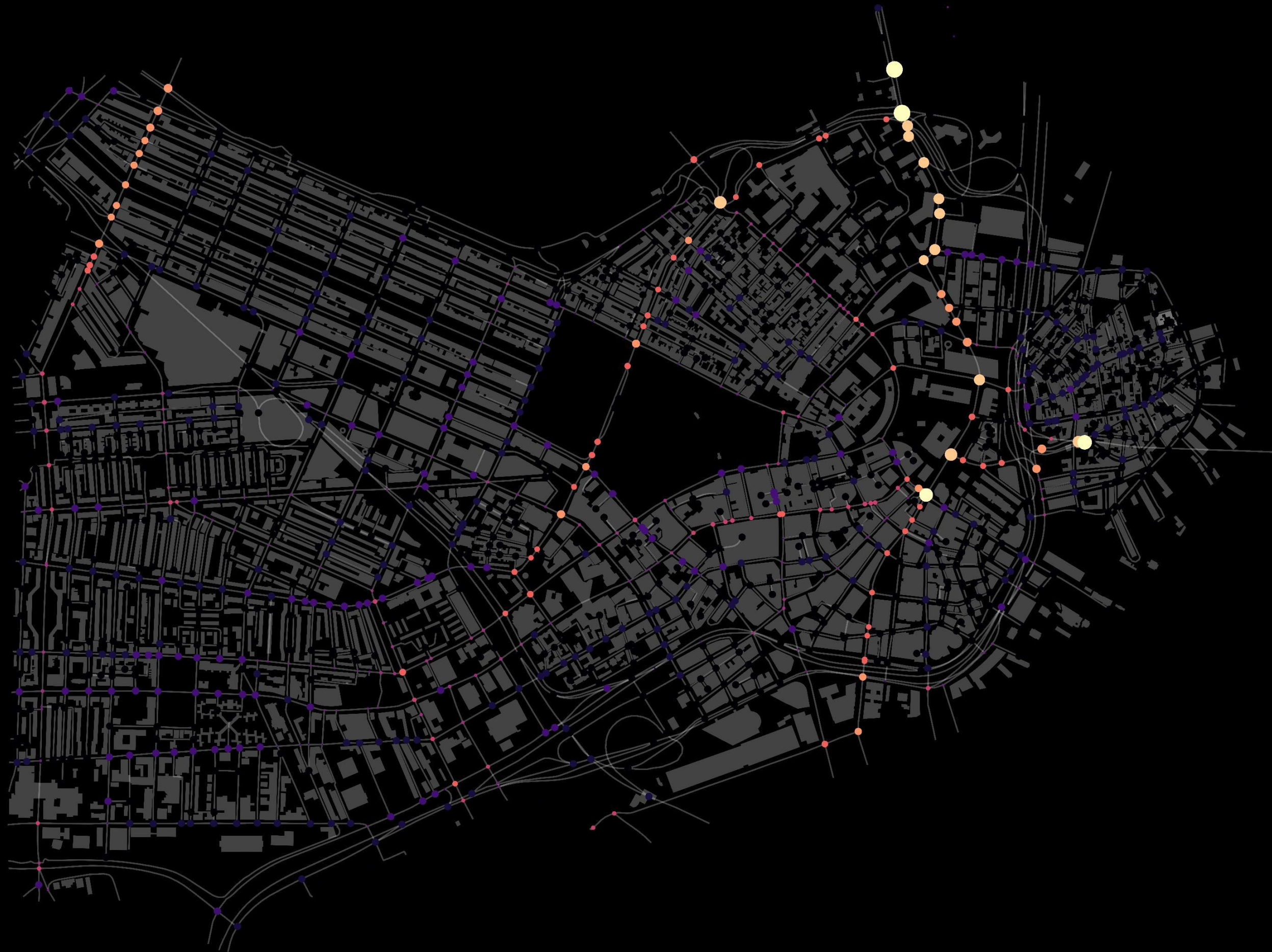


Centrality Measures

- Key measures for extracting the topological properties of the network.
- *Closeness centrality* measures to which extent a node i is near to all the other nodes.
- *Betweenness centrality* is based on the idea that a node is central if it lies between many other nodes, when it is traversed by many of the shortest paths connecting couples of nodes.

$$C_i^C = \frac{N - 1}{\sum_{j \in N, j \neq i} d_{ij}}$$

$$C_i^B = \sum_{j,k \in G, j \neq k \neq i} \frac{n_{jk}(i)}{n_{jk}}$$



Max



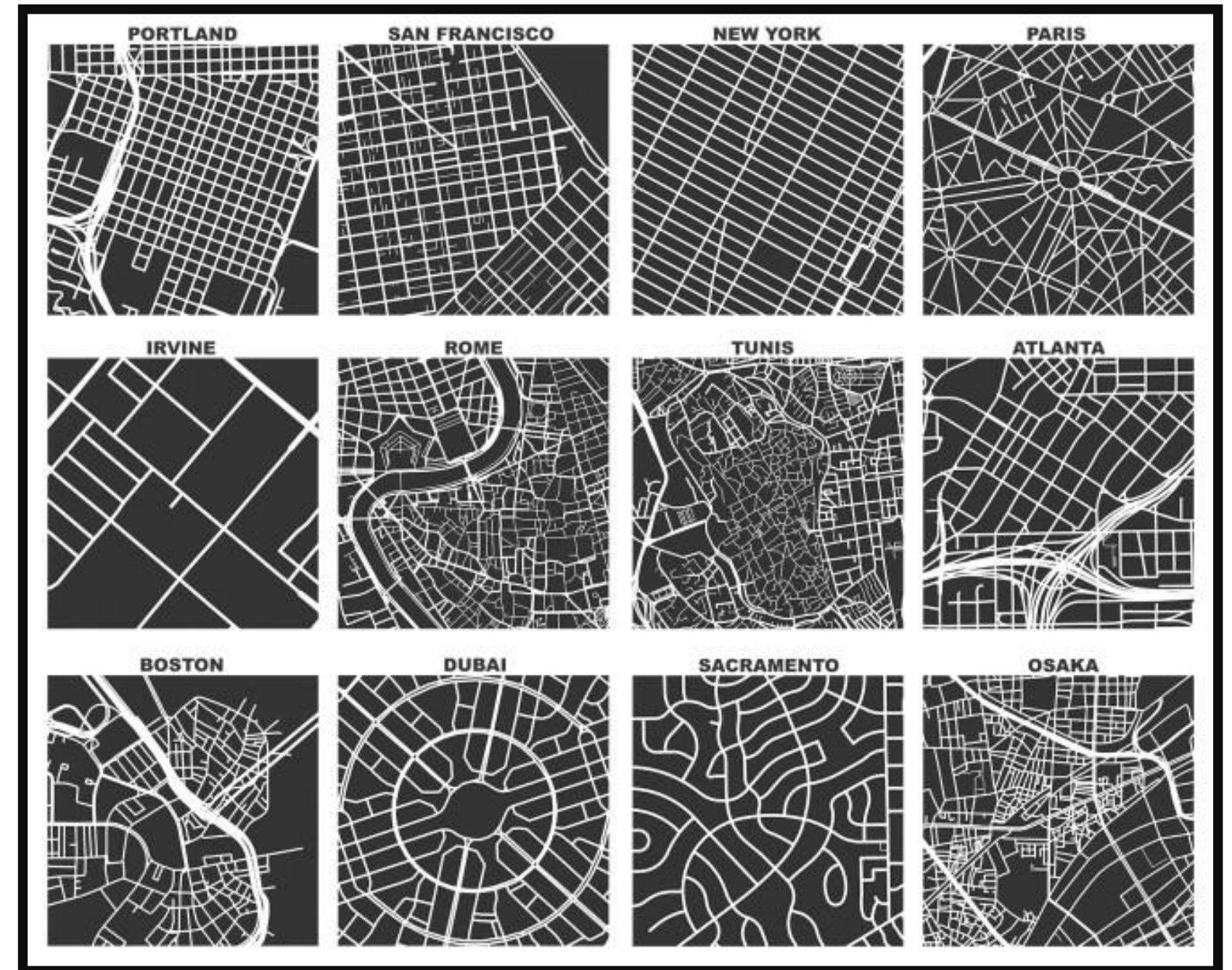
Min



Spatial Networks

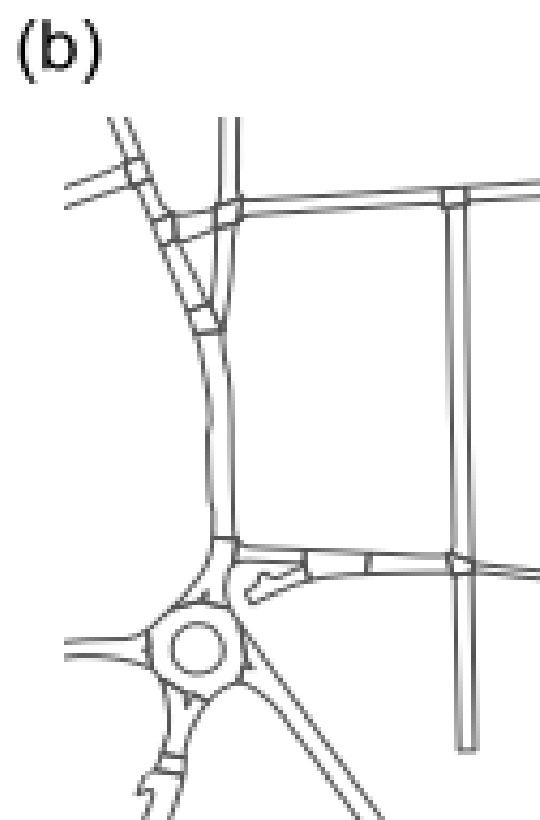
Network science and cities

- Cultural aspects of cities.
- Organic *vs* planned cities.
- Mental representation of space, mobility patterns and transport planning analysis.
- Services distribution and allocation, crime, segregation...

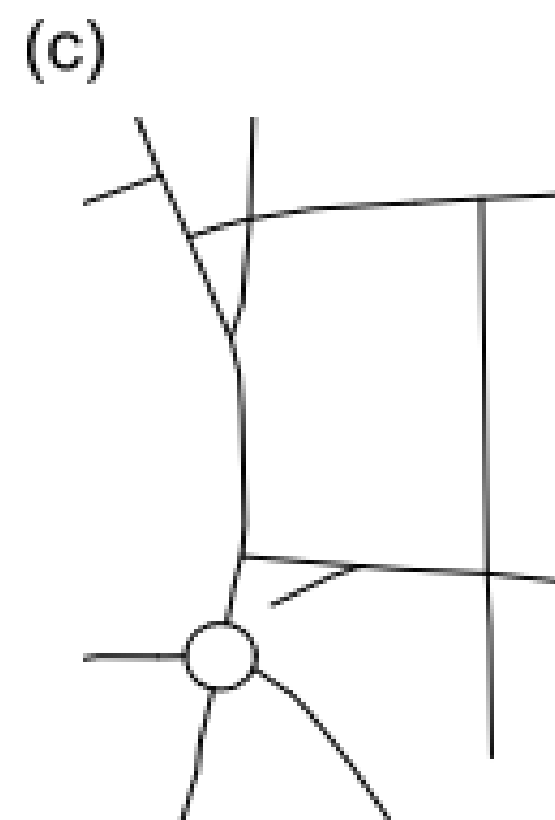




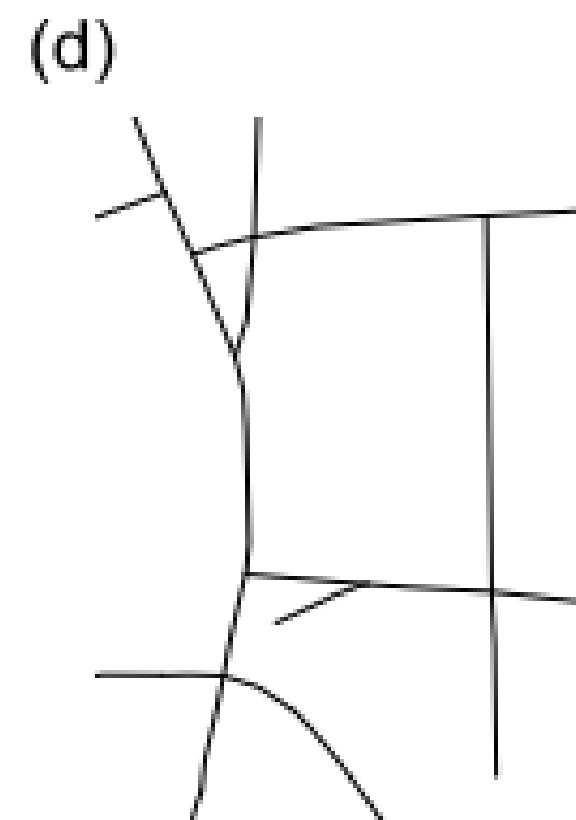
base map



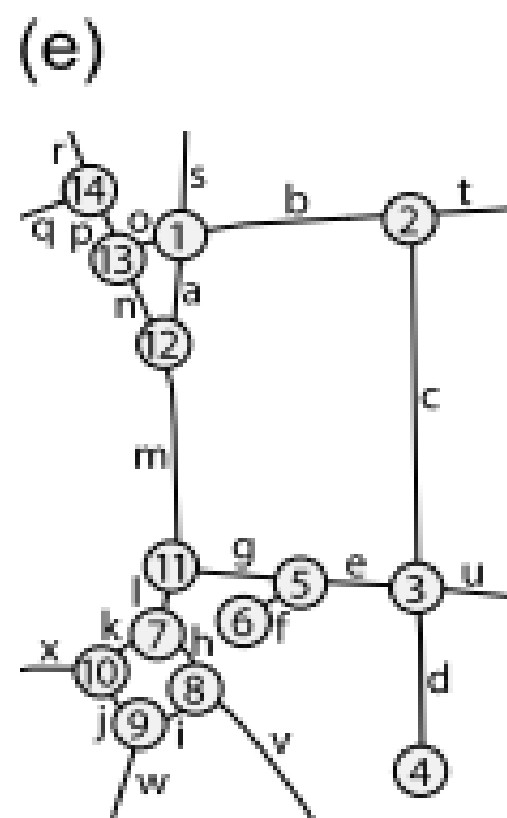
street surfaces /
polygonal partition



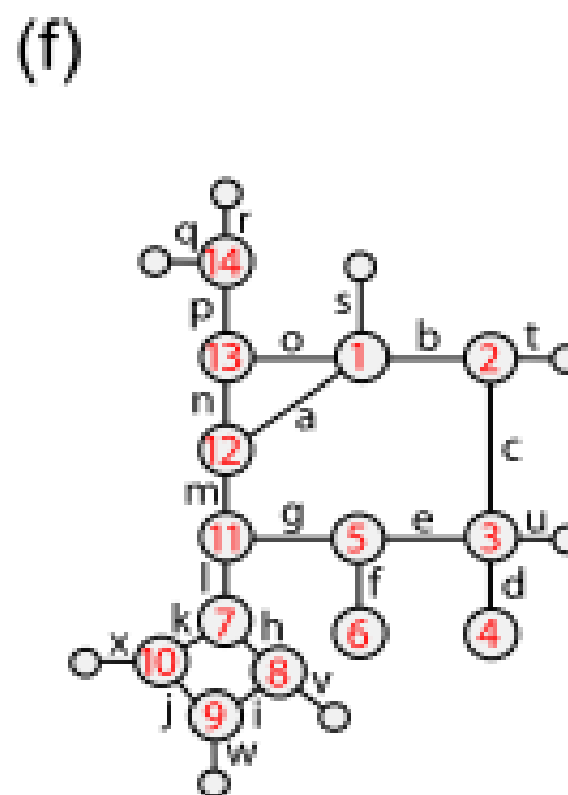
road centre line



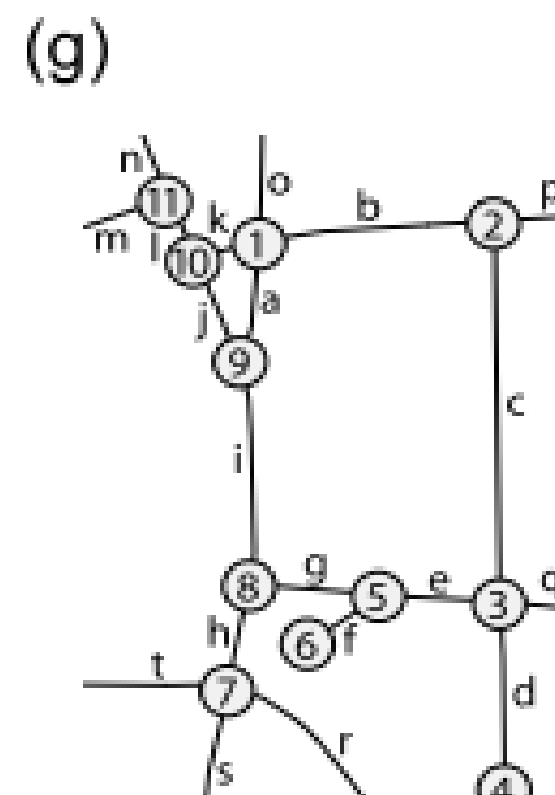
generalised road
centre line



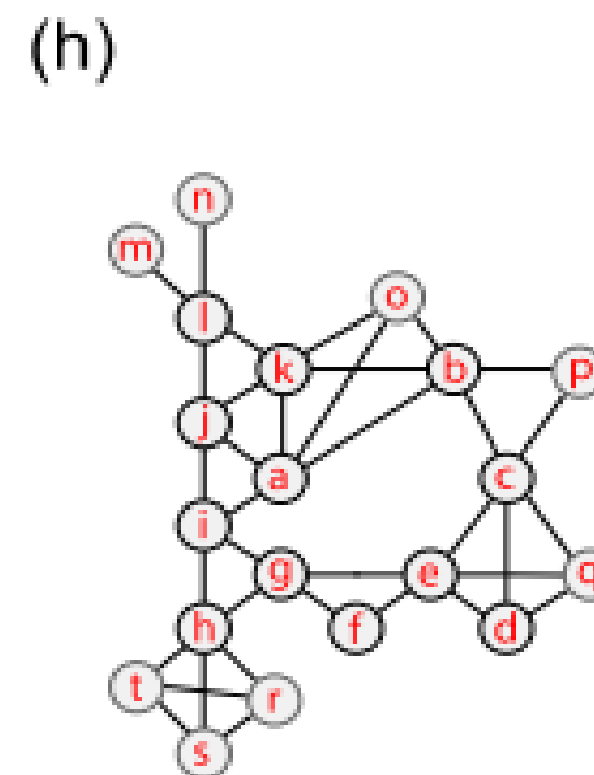
labelled nodes and links



junction graph
 $v = 14$ (22); $e = 24$
(including boundary vertices)



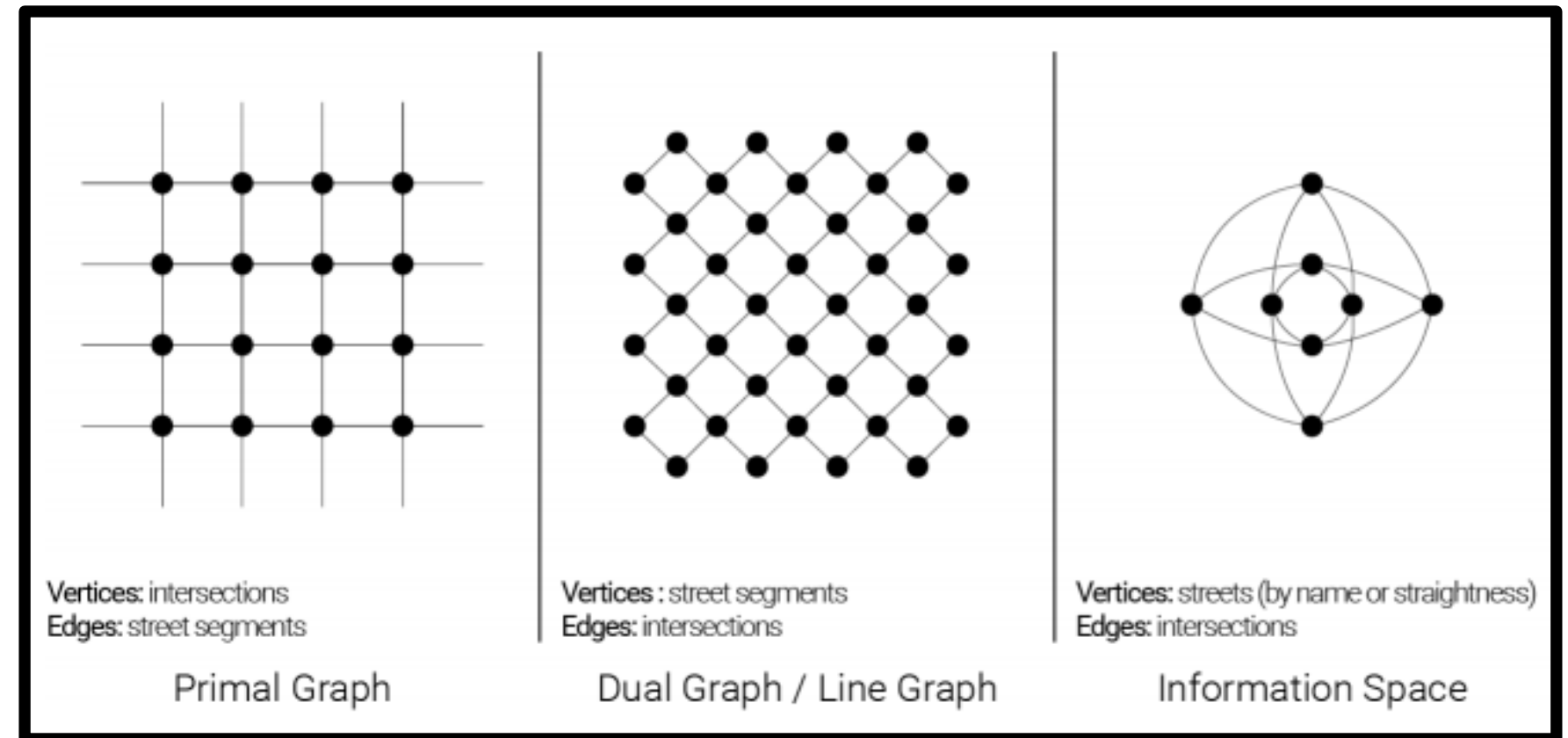
labelled generalised
nodes and links



street-segment graph
 $v = 12$ (20); $e = 19$ (35)
(including boundary segments)

Primal vs Dual representations

- Primal:
 - Node = street junction
 - Link = street segment
- Dual
 - Node = street segment (e.g. its centroid)
 - Links = they connect vertexes if the segments are linked in the original representation



Source Neira (2017)

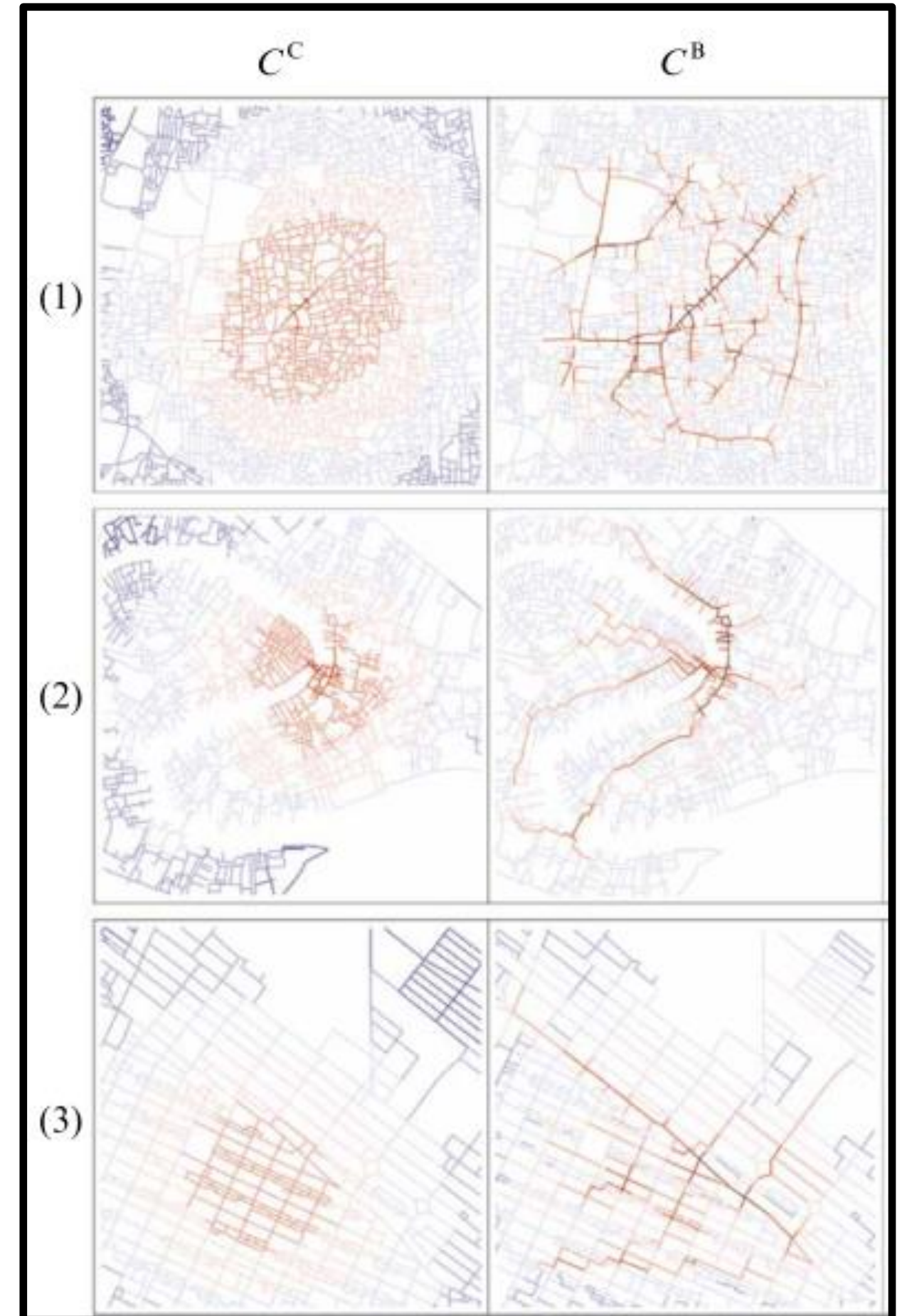
PG preserve geographic and metric information

Primal vs Dual graph



Centrality measures in cities

- Closeness centrality or betweenness centrality?
- What do they tell about the city?
- Connectivity and network resilience
- Drawbacks?



Python Tools

- *OSMnx* (street-network specific)
- *NetworkX*
- *iGraph* (faster than NetworkX)

OSMnx: Python for Street Networks

Check out the [journal article](#) about OSMnx.

OSMnx is a Python package for downloading administrative boundary shapes and street networks from OpenStreetMap. It allows you to easily construct, project, visualize, and analyze complex street networks in Python with NetworkX. You can get a city's or neighborhood's walking, driving, or biking network with a single line of Python code. Then you can simply visualize cul-de-sacs or one-way streets, plot shortest-path routes, or calculate stats like intersection density, average node connectivity, or betweenness centrality. You can download/cite the [paper](#) here.



In a single line of code, OSMnx lets you download, construct, and visualize the street network for, say, Modena Italy:

```
1 | import osmnx as ox
2 | ox.plot_graph(ox.graph_from_place('Modena, Italy'))
```

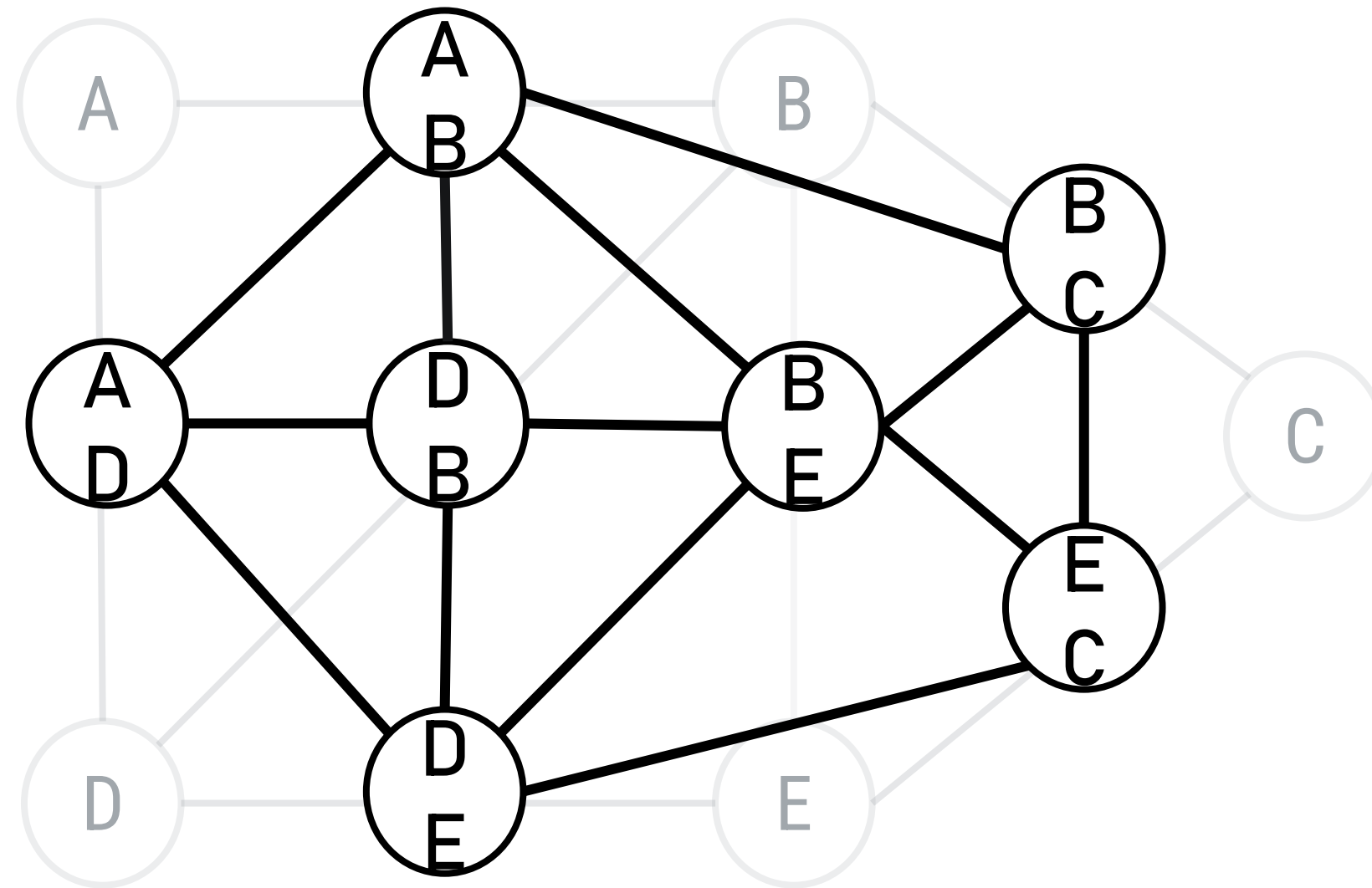


Sources and references

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- Porta, S, Crucitti, P. & Latora, V., 2006b. The Network Analysis Of Urban Streets: A Dual Approach. *Physica A* 369(2), pp.853–866.
- Barrington-Leigh, C, & Millard-Ball, A 2017. The world's user-generated road map is more than 80% complete. *PloS one* 12(8): e0180698.

Appendix

From Primal to Dual representation



DG allows to compute ANGULAR-CHANGE shortest-path

Computing deflection angles

